t-Product-Based Low-Rank Discriminative Graph Embedding for Robust Dimensionality Reduction of Hyperspectral Images

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***Abstract*—Graph embedding (GE) methods have shown promising performance in the dimensionality reduction (DR) of hyperspectral images (HSIs). However, overreliance on local information while neglecting global structure makes neighborhood graph construction highly susceptible to noise interference, and the projection operators in matrix form for DR may result in the loss of high-order structural information. To address these issues, we propose a t-product-based low-rank discriminative graph embedding (t-LRDGE) model. Specifically, we first integrate tensor-product (t-product) operations with a customized third-order tensor set representation to efficiently capture the intrinsic structure of hyperspectral data. Subsequently, we incorporate tensor low-rank decomposition into the GE for denoising, while considering the global and local information of the data to generate robust low-dimensional data. Finally, we introduce prior discriminative constraints on semantic information to further enhance the discriminative power of the model. In addition, we have developed an iterative algorithm based on the alternating direction method of multipliers (ADMMs) for solving t-LRDGE. Extensive experiments conducted on three datasets demonstrate that t-LRDGE outperforms existing state-of-the-art models in terms of both discriminative capability and robustness, effectively leveraging global and local information while preserving the intrinsic spatial–spectral structure. The source code is available at: https://github.com/loh0324/t-LRDGE**

***Index Terms*—Global–local information, graph embedding (GE), hyperspectral image (HSI), low-rank, tensor-product (t-product).**

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NOMENCLATURE

$\mathbf{X}$	Matrix.
$\mathcal{X}$	Third-order tensor.
$\mathcal{X}(:,i,j)$	(i,j)th mode-1 fiber (tube) of $\mathcal{X}$.
$\mathcal{X}(i,:,j)$	(i,j)th mode-2 fiber (tube) of $\mathcal{X}$.
$\mathcal{X}(i,j,:)$	(i,j)th mode-3 fiber (tube) of $\mathcal{X}$.
$\mathcal{X}(i,:,:)$	ith horizontal slice of $\mathcal{X}$.
$\mathcal{X}(:,j,:)=\mathcal{X}_{(j)}$	jth lateral slice of $\mathcal{X}$.
$\mathcal{X}(:,:,k)=\mathcal{X}^{(k)}$	kth frontal slice of $\mathcal{X}$.
$\|\cdot\|_1$, $\|\cdot\|_F$, $\|\cdot\|_*$	ℓ_1-norm, Frobenius-norm, and nuclear-norm.
$\widetilde{\mathcal{X}}$	Fourier transform of $\mathcal{X}$, where $\widetilde{\mathcal{X}}=\text{fft}(\mathcal{X})$.
$\overline{\mathbf{X}}$	Block-diagonal form of $\widetilde{\mathcal{X}}$.
$\mathcal{X}'$	Tensor (conjugate) transpose.
$\mathbf{X}^T$	Matrix (conjugate) transpose.

I. INTRODUCTION

HYPERSPECTRAL remote sensing can simultaneously acquire 2-D spatial and 1-D spectral information of ground objects as a cutting-edge Earth observation technology [1]. Hyperspectral images (HSIs) capture hundreds of contiguous narrow spectral bands and contain abundant land-cover information [2], which have been successfully applied in many fields such as urban management [3], medical imaging [4], and environmental monitoring [5]. However, the overwhelming spectral bands possess high spectral correlation and substantial redundancy, which consumes massive computational resources and causes "the curse of dimensionality" that affects the accuracy and effectiveness of high-dimensional data analysis [6]. Consequently, dimensionality reduction (DR) for HSI is essential for the subsequent HSI analysis.

Currently, the DR methods for HSI are mainly divided into two categories: feature selection (FS) and feature extraction (FE). FS [7] selects a representative subset from the original bands based on certain criteria to achieve DR. In contrast, FE [8] embeds high-dimensional data into a low-dimensional space by linear or nonlinear data mapping, and then obtains low-dimensional features that reflect the intrinsic information. This article focuses on FE methods to reduce the dimensionality of HSI.

Principal component analysis (PCA) [9] and linear discriminant analysis (LDA) [10] are widely used FE methods in

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DR. Among them, PCA aims to find a projection direction to maximize the variance of the data. LDA seeks the optimal discriminative subspace, where the ratio of distances between the interclass and the intraclass is maximized. These statistic-based linear methods can achieve some degree of data DR. However, these linear FE methods fail to exploit the nonlinear structures in HSI data. To solve this problem, manifold learning techniques were proposed. Nonlinear methods such as isometric feature mapping (ISOMAP) [11], locally linear embedding (LLE) [12], and Laplacian eigenmaps (LEs) [13] characterize the nonlinear manifold structure of the data from a local perspective. To enable LLE and LE in dealing with new samples [14], researchers have developed two linear approximation methods, neighborhood preserving embedding (NPE) [15] and locality preserving projection (LPP) [16].

To better unify these linear and nonlinear methods, graph embedding (GE) [17] has been developed to reveal and preserve similarities between samples via graph structures. Numerous graph-based methods have emerged early to deal with complex data, such as local Fisher discriminant analysis (LFDA) [18], supervised LPP (SLPP) [19], local neighborhood structure preserved embedding (LNSPE) [20], and local geometric structure Fisher analysis (LGSFA) [21]. Since then, researchers have further developed tensor-based GE methods to leverage the spatial–spectral information in HSI. Zheng et al. [22] applied the tensor LPP (TLPP) [23] method to the task of DR and classification of HSI, which effectively incorporates the GE framework by linearly approximating the eigenfunctions of the Laplace–Beltrami operator in the tensor space. Subsequently, a modified TLPP (MTLPP) [24] was proposed to construct the neighborhood graph with the region covariance descriptor. Feng and Wang [25] proposed a weighted MTLPP (WMTLPP). Although these GE methods have advanced in handling high-dimensional data, they employ a fixed number of neighbors for graph construction. These methods tend to overfocus on locality while neglecting the global structure, and are sensitive to noise. Therefore, it is difficult to fully meet the demand for DR of HSI data in complex scenes.

On the other hand, low-rank decomposition methods are effective in improving robustness to noise [26] and preserving global data structure. Candés et al. [27] proposed the robust PCA (RPCA), which decomposes the data matrix into a low-rank part and a sparse part. Liu et al. [28] developed a low-rank representation (LRR) model for clustering noisy data. Researchers further extended RPCA to the tensor domain and proposed tensor robust PCA (TRPCA) [29]. Although these methods validate the robustness of low-rank methods, they do not enable DR of high-dimensional data. To address this issue, Wang et al. [30] combined LRR and projection learning to build a low-rank embedding (LRE) integrated model and achieve robust FE. Deng et al. [31] proposed an $L_{2,p}$-norm-based robust embedding regression method ($L_{2,p}$-RER) for robust feature DR of HSI, which can effectively suppress the impact of noises and reduce the feature dimensions. Also, they extended it to the tensor field and presented a t-linear tensor subspace learning (tLTSL) [32] for FE of HSI based on t-product projection. The application of t-product-based operations has expanded into various fields. For instance, these operations have shown excellent performance compared with matrix-based methods in image restoration [33] and clustering [34] fields. While methods like tLTSL [32] successfully utilize low-rank constraints for global denoising, they overlook local manifold structures and discriminative semantic information.

Recently, deep learning (DL)-based DR has also been investigated for HSI analysis due to their strong nonlinear representation capability. Representative studies include DLPNet [35] and DMRNet [36], which incorporate manifold-related constraints into deep networks to learn compact and discriminative features. However, despite their promising performance, DL models often rely on substantial labeled data and careful hyperparameter tuning, and their learned transformations are less interpretable. In general, existing GE methods suffer from noise sensitivity and insufficient utilization of global information, while low-rank methods improve robustness but may sacrifice local manifold details. DL-based methods can introduce high training overhead and require substantial labeled data.

To address these issues, we propose the a t-product-based low-rank discriminative graph embedding (t-LRDGE) model, which focuses on projection learning based on a tensor, considering both global and local information of HSIs to achieve robust and efficient FE. In combination with the tensor-product (t-product) operation and third-order tensor set representation, the t-LRDGE model can efficiently represent and transform tensors without destroying spatial structure. In addition, the t-LRDGE model integrates three key components, i.e., global structure, local geometric features, and semantic priors of HSI data, into a unified framework for joint optimization. The main contributions of this article are summarized as follows.

1) We propose the t-LRDGE model constructed in a tensor space to address DR for HSI. By leveraging t-product operations and third-order tensor set representations, this approach efficiently captures rich spatial–spectral information and transforms it without destroying the intrinsic spatial structure of the data.

2) The t-LRDGE model extracts global–local structural features by jointly optimizing two proposed modules: a local-aware GE (LGE) module using covariance descriptors and the Earth mover's distance (EMD) to characterize local neighborhoods, and a global-preserving low-rank (GLR) module that preserves global structure and low-rank properties to suppress noise.

3) We enhance the discriminative capability of the model by incorporating prior semantic information as discriminative constraints. This strategy establishes a mapping between features and labels, ensuring that class-discriminative information of data is maintained within the projected low-dimensional subspace.

4) We develop an efficient iterative optimization algorithm to solve the proposed t-LRDGE model. Extensive comparative experiments on three datasets demonstrate that our method outperforms existing state-of-the-art methods, exhibiting superior discriminability and noise robustness.

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II. Notations and Preliminaries

A. Notations

In this section, we introduce the tensor representation and the t-product-based operations [37], [38], [39], [40]. Throughout this article, uppercase letters and Euler script letters represent matrices and third-order tensors, e.g., $\mathbf{X}$ and $\mathcal{X}$. The conjugate transpose of matrix $\mathbf{X}$ and tensor $\mathcal{X}$ are denoted as $\mathbf{X}^T$ and $\mathcal{X}'$. The ℓ_1-norm, the Frobenius norm, and the nuclear norm of the tensor $\mathcal{X}$ are represented by $\|\mathcal{X}\|_1$, $\|\mathcal{X}\|_F$, and $\|\mathcal{X}\|_*$, respectively. For a third-order tensor $\mathcal{X} \in \mathbb{R}^{n_1\times n_2\times n_3}$, $\mathcal{X}(:,i,j) \in \mathbb{R}^{n_1\times 1\times 1}$, $\mathcal{X}(i,:,j)$, and $\mathcal{X}(i,j,:)$ represent the (i,j)th mode-1, mode-2, and mode-3 fiber (tube), respectively. $\mathcal{X}(i,:,:)$, $\mathcal{X}(:,j,:)$, and $\mathcal{X}(:,:,k)$ are the horizontal, lateral, and frontal slices, respectively. For simplicity, $\mathcal{X}(:,j,:)$ and $\mathcal{X}(:,:,k)$ can be written as $\mathcal{X}_{(j)}$ and $\mathcal{X}^{(k)}$. $\widetilde{\mathcal{X}} \in \mathbb{C}^{n_1\times n_2\times n_3}$ is obtained by applying Fast Fourier Transform (FFT) on $\mathcal{X}$ along its third direction, with $\widetilde{\mathcal{X}} = \text{fft}(\mathcal{X})$. The inverse operation $\mathcal{X} = \text{ifft}(\widetilde{\mathcal{X}})$ recovers the original tensor. The notations are summarized in Nomenclature.

Next, we introduce the definitions associated with the t-product.

Definition 1 (t-product): Given $\mathcal{X} \in \mathbb{R}^{n_1\times n_2\times n_3}$ and $\mathcal{Y} \in \mathbb{R}^{n_2\times n_4\times n_3}$, the t-product $\mathcal{X} * \mathcal{Y}$ yields a tensor $\mathcal{Z}$ of size $n_1 \times n_4 \times n_3$, and its specific definition is as follows:

$$\mathcal{Z} = \mathcal{X} * \mathcal{Y} = \text{fold}\left(\circ(\mathcal{X}) \cdot \text{unfold}(\mathcal{Y})\right) \tag{1}$$

where the block circulant operator is defined as

$$\circ(\mathcal{X}) = \begin{bmatrix} \mathcal{X}^{(1)} & \mathcal{X}^{(n_3)} & \cdots & \mathcal{X}^{(2)} \\ \mathcal{X}^{(2)} & \mathcal{X}^{(1)} & \ddots & \vdots \\ \vdots & \vdots & \ddots & \mathcal{X}^{(n_3)} \\ \mathcal{X}^{(n_3)} & \mathcal{X}^{(n_3-1)} & \cdots & \mathcal{X}^{(1)} \end{bmatrix} \in \mathbb{R}^{n_1 n_3 \times n_2 n_3} \tag{2}$$

and the block vectorizing operator, together with its inverse operator, are specified as

$$\text{unfold}(\mathcal{X}) = \begin{bmatrix} \mathcal{X}^{(1)} \\ \mathcal{X}^{(2)} \\ \vdots \\ \mathcal{X}^{(n_3)} \end{bmatrix} \in \mathbb{R}^{n_1 n_3 \times n_2}, \quad \text{unfold}(\text{fold}(\mathcal{X})) = \mathcal{X}. \tag{3}$$

Using the t-product definition, we can naturally derive several practical definitions and properties. Let $\widetilde{\mathcal{X}}$ be the tensor generated by the FFT of all tubes along the third dimension of $\mathcal{X}$. Next, we denote the block-diagonal form of $\widetilde{\mathcal{X}}$ in the Fourier domain as $\overline{\mathbf{X}}$, which is written as

$$\begin{aligned} \overline{\mathbf{X}} &= \text{blockdiag}\left(\text{fft}(\mathcal{X})\right) = \text{blockdiag}\left(\widetilde{\mathcal{X}}\right) \\ &= \begin{bmatrix} \widetilde{\mathcal{X}}^{(1)} & & & \\ & \widetilde{\mathcal{X}}^{(2)} & & \\ & & \ddots & \\ & & & \widetilde{\mathcal{X}}^{(n_3)} \end{bmatrix} \in \mathbb{R}^{n_1 n_3 \times n_2 n_3}. \end{aligned} \tag{4}$$

It is well known that at this point, the complex tensor computation in the original domain is transformed into a simpler matrix multiplication in the Fourier domain.

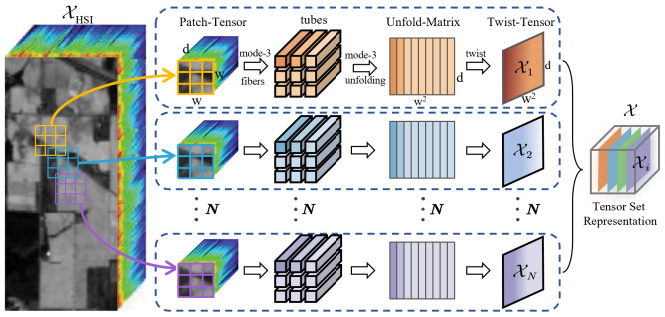


Fig. 1. Flowchart of tensor set representation construction. First, a $w \times w$ sliding window extracts local patch-tensors from the $\mathcal{X}_{\text{HSI}}$. Second, each patch-tensor undergoes mode-3 fiber extraction and mode-3 unfolding to yield $d \times w^2$ unfold-matrices, which are then converted into $d \times 1 \times w^2$ twist-tensors via the twist operator. Finally, stacking these tensors along mode-2 constructs the final tensor set representation $\mathcal{X} \in \mathbb{R}^{d\times N\times w^2}$.

Property 1: The t-product in the original domain is equivalent to the matrix multiplication in the Fourier domain

$$\mathcal{Z} = \mathcal{X} * \mathcal{Y} \Leftrightarrow \overline{\mathbf{Z}} = \overline{\mathbf{X}}\,\overline{\mathbf{Y}}. \tag{5}$$

Definition 2 (Identity Tensor): An identity tensor $\mathcal{I} \in \mathbb{R}^{n_1\times n_1\times n_3}$ has the $n_1 \times n_1$ identity matrix as its first frontal slice, with all other frontal slices being zero.

Definition 3 (Orthogonal Tensor): A tensor $\mathcal{X} \in \mathbb{R}^{n_1\times n_1\times n_3}$ is classified as orthogonal tensor if $\mathcal{X} * \mathcal{X}' = \mathcal{X}' * \mathcal{X} = \mathcal{I}$, with $\mathcal{I}$ being the identity tensor.

Definition 4 (Tensor Conjugate Transpose): For $\mathcal{X} \in \mathbb{R}^{n_1\times n_2\times n_3}$, the conjugate transpose $\mathcal{X}'$ is an $n_2 \times n_1 \times n_3$ tensor, derived by transposing each frontal slice of $\mathcal{X}$ and subsequently inverting the sequence of the transposed frontal slices from two to n_3.

Definition 5 (f-diagonal Tensor): A tensor is referred to as an f-diagonal tensor if every frontal slice of it is a diagonal matrix.

Definition 6 (Tensor Singular Value Decomposition (t-SVD) [37]): Given an $n_1 \times n_2 \times n_3$ tensor $\mathcal{X}$, its Singular Value Decomposition (t-SVD) is given by

$$\mathcal{X} = \mathcal{U} * \mathcal{S} * \mathcal{V}^T \tag{6}$$

where $\mathcal{U} \in \mathbb{R}^{n_1\times n_1\times n_3}$ and $\mathcal{V} \in \mathbb{R}^{n_2\times n_2\times n_3}$ are orthogonal tensors, $\mathcal{S} \in \mathbb{R}^{n_1\times n_2\times n_3}$ is f-diagonal tensor, and * represents the t-product.

B. Tensor Set Representation

Below, we describe the tensor set representation constructed in this article. The entire process is shown in Fig. 1. First, to capture the spatial–spectral information of HSI, a $w \times w$ sliding window is applied to the $\mathcal{X}_{\text{HSI}} \in \mathbb{R}^{m\times n\times d}$ (where $m \times n$ represents the spatial dimension, and d is the spectral dimension). This process generates a patch-tensor set $\{\mathcal{X}_{w_1}, \mathcal{X}_{w_2}, \ldots, \mathcal{X}_{w_N}\} \in \mathbb{R}^{w\times w\times d}$, where $\mathcal{X}_{w_i}$ represents the local neighborhood around the central pixel within the sliding window. This local neighborhood can be regarded as a spatial–spectral tensor representation, ultimately forming N samples with integrated spatial–spectral information. Second,

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each patch tensor $\mathcal{X}_{w_i}$ extracts mode-3 fibers (tubes), and then these tubes are unfolded into a matrix $\mathbf{X}_i \in \mathbb{R}^{d\times w^2}$ through mode-3 unfolding. Finally, we apply the twist operator to transform each matrix $\mathbf{X}_i$ into a third-order tensor, i.e., $\text{twist}(\mathbf{X}_i) = \mathcal{X}_i \in \mathbb{R}^{d\times 1\times w^2}$. To obtain the tensor set representation, each of the third-order tensor samples after twisting is stacked along the second direction to form the tensor set representation $\mathcal{X} \in \mathbb{R}^{d\times N\times w^2} = \{\mathcal{X}_i\}_{i=1}^N$. All subsequent inputs and operations are based on this tensor set representation.

III. Proposed Method

In this section, we first introduce the proposed modules designed to handle specific issues. Subsequently, we establish the t-LRDGE projection learning model based on the proposed modules. Finally, an iterative algorithm is derived within the framework of the alternating direction method of multipliers (ADMMs) [41] to solve the optimization t-LRDGE model.

A. Modules

In HSI DR processing, a high-performance projection learning model should account for spatial–spectral structures while exhibiting robustness to damage. In addition, it should preserve the discriminative ability and intrinsic relationships within the HSI data. Therefore, we propose three key modules, namely, the LGE module, the GLR module, and the discriminative enhancement (DE) module, and formulate them using t-product-based operations. We integrate these modules into a unified joint optimization framework. This enables the GLR constraint, local geometric information, and semantic constraints to mutually refine each other during the learning process.

1) LGE Module: In HSI processing, the local geometric structure of samples (e.g., similarity and neighborhood relationships) plays a crucial role in uncovering the underlying distribution patterns of the data. Especially in high-dimensional spaces, local information provides a more precise representation of the data's intrinsic manifold structure. To leverage this, we propose an LGE module that builds upon the theoretical framework of LPP [16] and preserves local structures by constructing neighborhood graphs between samples. Specifically, this module first implements a t-product-based LPP GE method and improves the strategy of constructing the adjacency graph G. It can effectively maintain local relationships and ensure that key information remains intact in the low-dimensional representation during the DR process.

During the graph construction phase, diverging from the Euclidean distance used in methods like TLPP [23], we employ covariance descriptors combined with the EMD. Specifically, region covariance focuses on characterizing the statistical correlations among spectral bands, while EMD captures the discrepancies in global distribution patterns of pixel spectra. Combining them ensures that the constructed graph encompasses both statistical and distributional similarities. The flow of this LGE module can be presented as follows.

1) *Step 1:* Let $\mathbf{x}_t \in \mathbb{R}^d$ $(t = 1, 2, \ldots, w^2)$ denote the spectral vectors situated within a local window centered at the ith pixel, and let $\bar{\mathbf{x}}_i = (1/w^2)\sum_{t=1}^{w^2} \mathbf{x}_t$ be the corresponding mean vector. To construct the adjacency graph, we characterize each sample $\mathcal{X}_i \in \mathbb{R}^{d\times 1\times w^2}$ using the regional covariance feature $\mathbf{C}_i$. The covariance matrix $\mathbf{C}_i$ of $\mathcal{X}_i$ is computed as follows:

$$\mathbf{C}_i = \frac{1}{w^2-1}\sum_{t=1}^{w^2}(\mathbf{x}_t - \bar{\mathbf{x}}_i)(\mathbf{x}_t - \bar{\mathbf{x}}_i)^T. \tag{7}$$

Given that the covariance descriptors lie on a Riemannian manifold [42], we employ the log-Euclidean distance to assess the similarity between $\mathbf{C}_i$ and $\mathbf{C}_j$. Consequently, the distance between the corresponding samples $\mathcal{X}_i$ and $\mathcal{X}_j$ is determined by this metric, which is formulated as

$$\mathrm{D}_{\mathrm{RC}}(\mathcal{X}_i, \mathcal{X}_j) = \left\|\log(\mathbf{C}_i) - \log(\mathbf{C}_j)\right\|_F. \tag{8}$$

Then, different from the regional covariance that characterizes the statistical correlations between different bands, we further introduce EMD to describe the differences in the distribution patterns of pixel spectra within local regions. This is achieved by constructing spectral value histograms

$$\mathrm{D}_{\mathrm{EMD}}(\mathcal{X}_i, \mathcal{X}_j) = \text{Wasserstein}(T_i, T_j) \tag{9}$$

where T_i and T_j denote the spectral histograms corresponding to $\mathcal{X}_i$ and $\mathcal{X}_j$, respectively. To quantify the similarity between these two probability distributions, we employ the Wasserstein distance, commonly known as the EMD.

We take a weighted linear fusion strategy to fuse the two, and use the merged distance $\mathrm{D}(\mathcal{X}_i, \mathcal{X}_j)$ to construct the adjacency graph G for tensor set representation. Specifically, this process is formulated as

$$\begin{aligned}\mathrm{D}(\mathcal{X}_i, \mathcal{X}_j) &= \alpha\mathrm{D}_{\mathrm{RC}}(\mathcal{X}_i, \mathcal{X}_j) + \beta\mathrm{D}_{\mathrm{EMD}}(\mathcal{X}_i, \mathcal{X}_j)\\ &\quad i, j = 1, 2, \ldots, N\end{aligned} \tag{10}$$

where α and β are the weighting parameters satisfying $\alpha + \beta = 1$. Subsequently, an adjacency graph G consisting of N nodes is established via the k-nearest neighbors (kNNs) method.

2) *Step 2:* To construct the adjacency graph, we calculate the edge weights $\{W_{ij}\}_{i,j=1}^N$. In this study, the heat kernel is adopted to measure the similarity between nodes $\mathcal{X}_i$ and $\mathcal{X}_j$, formulated as

$$W_{ij} = \begin{cases}\exp\left(-\dfrac{\mathrm{D}(\mathcal{X}_i, \mathcal{X}_j)^2}{2\sigma^2}\right), & \text{if } \mathcal{X}_i \in \Omega_k(\mathcal{X}_j)\\ 0, & \mathcal{X}_j \notin \Omega_k(\mathcal{X}_i)\end{cases} \tag{11}$$

where $\Omega_k(\mathcal{X}_i)$ represents the set of kNNs for $\mathcal{X}_i$, and this set is obtained through the fused distance $\mathrm{D}(\cdot,\cdot)$. σ serves as the kernel width parameter. Consequently, a sparse weight matrix $\mathbf{W} \in \mathbb{R}^{N\times N}$ is derived.

3) *Step 3:* Denote $\mathcal{P}$ as projection tensor that maps high-dimensional data $\mathcal{X}_i$ into a low-dimensional space. By minimizing the distances between low-dimensional samples, we can learn the projection tensor $\mathcal{P}$. To derive

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the optimal projection tensor, we solve the optimization problem formulated as

$$\min_{\mathcal{P}} \sum_{i=1}^{N}\sum_{\substack{j=1\\ i\neq j}}^{N} \left\|\mathcal{P}*\mathcal{X}_i-\mathcal{P}*\mathcal{X}_j\right\|_F^2 \cdot W_{i,j}$$
$$\text{s.t.} \sum_i \|\mathcal{P}*\mathcal{X}_i\|_F^2 \cdot d_{ii}=1 \tag{12}$$

where $\mathcal{X}_i$ and $\mathcal{X}_j \in \mathbb{R}^{d\times 1\times w^2}$ are the ith and jth tensor samples in tensor set $\mathcal{X}$, $\mathcal{P}\in\mathbb{R}^{r\times d\times w^2}$ is the projection tensor, r ($r \ll d$) is the projected subspace dimension, W_{ij} presents the similarity between $\mathcal{X}_i$ and $\mathcal{X}_j$ as defined in (11), and $d_{ii}=\sum_j W_{ij}$.

2) Global-Preserving Low-Rank Module: Although (12) effectively preserves data locality, it neglects global information during DR and struggles to handle noise interference. These two factors directly impact the DR outcome, making it difficult to fully meet practical application requirements.

To address this issue, we introduce a GLR module based on tLTSL [32] to preserve global structure and reduce noise pollution. This module aims to decompose data into a low-rank part and a sparse part via tensor low-rank decomposition, thereby effectively removing noise and preventing information loss. Distinct from global models like tLTSL or TRPCA [29] that process the entire HSI, we construct the input tensor using only training samples. This strategy avoids interference from redundant background information and focuses on extracting discriminative features from a concise sample set. The mathematical formulation of the objective function for this module is given by

$$\min_{\mathcal{L},\mathcal{E},\mathcal{P}} \text{rank}(\mathcal{L})+\lambda\|\mathcal{E}\|_1$$
$$\text{s.t.}\quad \mathcal{P}*\mathcal{X}=\mathcal{P}*\mathcal{L}+\mathcal{E},\quad \mathcal{P}'*\mathcal{P}=\mathcal{I} \tag{13}$$

where $\mathcal{P}*\mathcal{X}\in\mathbb{R}^{r\times N\times w^2}$ represents the projected low-dimensional data of the noisy observation tensor set, which is further separated into a low-rank part $\mathcal{L}\in\mathbb{R}^{d\times N\times w^2}$ and a sparse part $\mathcal{E}\in\mathbb{R}^{r\times N\times w^2}$ in the low-dimensional space. $\mathcal{I}\in\mathbb{R}^{d\times d\times w^2}$ represents the identity tensor.

To solve the constructed module (13), the tensor nuclear norm (TNN) [43] is first employed to approximate the rank of $\mathcal{L}$ to avoid the NP-hard problem of tensor rank minimizing. Correspondingly, this module can be rewritten as

$$\min_{\mathcal{L},\mathcal{E},\mathcal{P}} \|\mathcal{L}\|_*+\lambda\|\mathcal{E}\|_1$$
$$\text{s.t.}\quad \mathcal{P}*\mathcal{X}=\mathcal{P}*\mathcal{L}+\mathcal{E},\quad \mathcal{P}'*\mathcal{P}=\mathcal{I}. \tag{14}$$

3) DE Module: In the FE of HSI, incorporating prior label knowledge is paramount for augmenting the discriminative capability of features. However, the supervisory information of samples is not fully utilized in (12) and (13), which may lead to the projected data failing to maintain satisfactory discriminative ability in the low-dimensional space.

To solve this issue, we propose a DE module to leverage the mapping relationship between the low-dimensional subspace and the label space, rather than between the raw data space and the label space, thereby embedding strong label discriminability in the low-dimensional space. We define a parameter tensor $\mathcal{H}\in\mathbb{R}^{c\times r\times w^2}$ to establish a mapping between the low-dimensional data $\mathcal{P}*\mathcal{X}$ and the label tensor $\mathcal{Y}\in\mathbb{R}^{c\times N\times w^2}$, where c is the number of classes and N is the number of training samples. Specifically, $\mathcal{Y}$ is initialized as a zero tensor. For the ith sample with label c_i, we set $\mathcal{Y}(c_i,i,:)$ to 1. This operation is applied to all N samples, yielding the final label tensor $\mathcal{Y}$. This parameter tensor $\mathcal{H}$ is obtained by minimizing the classification error as follows:

$$\min_{\mathcal{H},\mathcal{P}} \sum_{i=1}^{N} l(\mathcal{Y}_i,\phi(\mathcal{P}*\mathcal{X}_i,\mathcal{H})) \tag{15}$$

where $\mathcal{Y}_i\in\mathbb{R}^{c\times 1\times w^2}$ is the label tensor of the ith sample $\mathcal{X}_i$, $\phi(\cdot)$ is a feature extractor based on the t-product, defined as $\phi(\text{input},\mathcal{H})=\mathcal{H}*\text{input}$, and $l(\cdot)=\|\|_F^2$ denotes the loss function of the classifier calculated over the training set. Equation (15) is equivalent to

$$\min_{\mathcal{H},\mathcal{P}} \|\mathcal{Y}-\mathcal{H}*\mathcal{P}*\mathcal{X}\|_F^2. \tag{16}$$

B. Model Formulation

We propose an integrated framework called t-LRDGE that combines the LGE module, the GLR module, and the DE module. This framework effectively addresses three key issues in DR of HSI: preserving both local and global structures, removing noise, and fully utilizing supervisory information. Ultimately, this leads to improved performance in DR, denoising, and classification. The t-LRDGE model can be expressed as

$$\min_{\mathcal{L},\mathcal{E},\mathcal{P},\mathcal{H}} \underbrace{\|\mathcal{L}\|_*+\lambda_1\|\mathcal{E}\|_2}_{\text{Global-Preserve Low-Rank Module}} + \underbrace{\lambda_2\|\mathcal{Y}-\mathcal{H}*\mathcal{P}*\mathcal{X}\|_F^2}_{\text{Discriminative Enhancement Module}}$$
$$+\underbrace{\lambda_3\sum_i\sum_j \left\|\mathcal{P}*\mathcal{X}_i-\mathcal{P}*\mathcal{X}_j\right\|_F^2 W_{ij}}_{\text{Local-Aware Graph Embedding Module}}$$
$$\text{s.t. } \mathcal{P}*\mathcal{X}=\mathcal{P}*\mathcal{L}+\mathcal{E},\quad \sum_i \|\mathcal{P}*\mathcal{X}_i\|_F^2 d_{ij}=1 \tag{17}$$

where $\mathcal{X}$, $\mathcal{P}$, $\mathcal{L}$, $\mathcal{E}$, $\mathcal{H}$, $\mathcal{Y}$, and W have the same definitions as in (12), (13), and (16), and λ_1, λ_2, and λ_3 serve as the tradeoff parameters to balance the weights of the three modules.

For the ease of understanding, the primary structure of the t-LRDGE model is illustrated in Fig. 2. As shown in this figure, the t-LRDGE model projects the high-dimensional data $\mathcal{X}$ into a low-dimensional representation using the learnable projection tensor $\mathcal{P}$. Furthermore, the noisy subspace representation $\mathcal{P}*\mathcal{X}$ retains the clean representation $\mathcal{P}*\mathcal{L}$ through a low-rank decomposition. In addition, the t-LRDGE model connects the t-linear subspace representations $\mathcal{P}*\mathcal{X}$ with the label tensor $\mathcal{Y}$ via another learnable projection tensor $\mathcal{H}$. The GE module embeds the affinity graph of tensor samples. As a conclusion, the mutual balance maintained between the three proposed modules helps the t-LRDGE model preserve more valuable information in the learned representations and reach the optimal solution.

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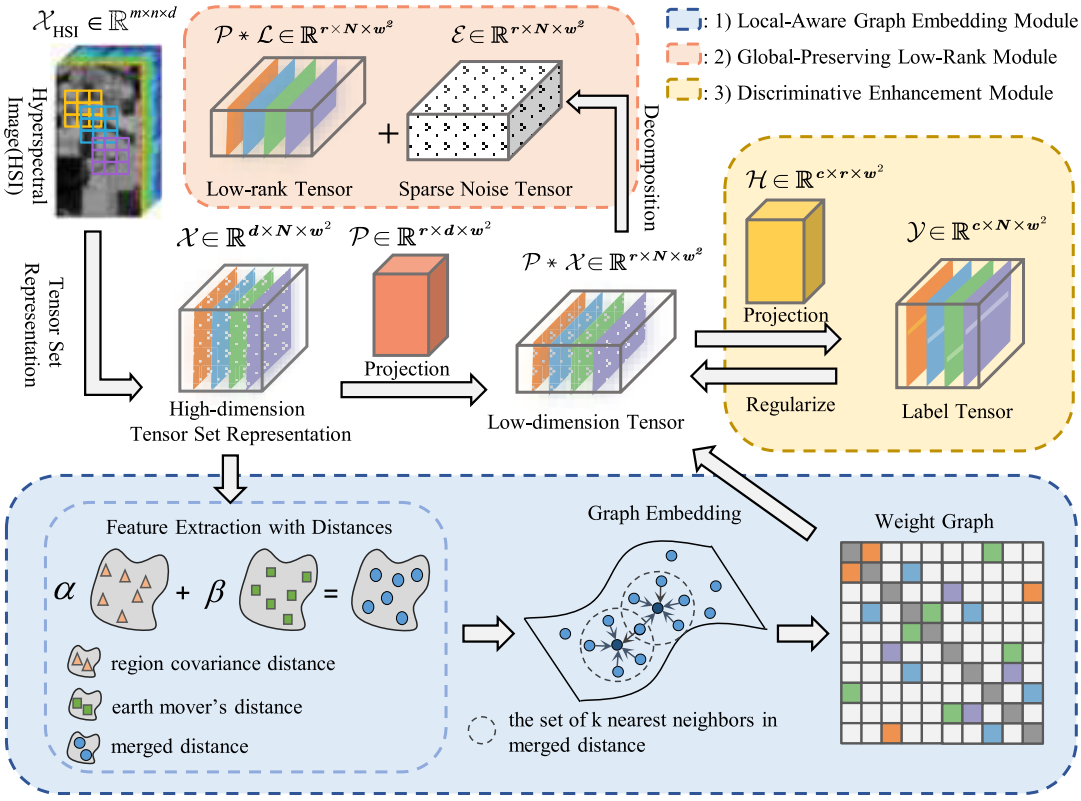


Fig. 2. Overall framework of the proposed t-LRDGE model consists of three modules. The balance among these modules allows the t-LRDGE model to achieve optimal performance. 1) LGE module extracts local structural information from high-dimensional data $\mathcal{X}$ (tensor set representation constructed in Section II-B) and embeds it into a low-dimensional subspace. 2) Global-preserving low-rank module decomposes low-dimensional data into a low-rank clean data part $\mathcal{P} * \mathcal{L}$ and a sparse part $\mathcal{E}$. 3) DE module introduces labels as supervisory information and strengthens feature discriminability by narrowing the gap between the low-dimensional space and the semantic label space $\mathcal{Y}$.

C. Model Optimization

In this section, we develop an iterative optimization algorithm to solve the t-LRDGE model utilizing the ADMMs [41]. To simplify the optimization problem, let $\mathcal{L} = \mathcal{M}$, $\mathcal{P} = \mathcal{Q}$, $\mathcal{H} * \mathcal{P} = \mathcal{G}$. Consequently, the optimization problem (17) can be converted to

$$\begin{aligned}\min_{\mathcal{M},\mathcal{L},\mathcal{E},\mathcal{H},\mathcal{G},\mathcal{P},\mathcal{Q}} \quad & \|\mathcal{M}\|_* + \lambda_1 \|\mathcal{E}\|_2 + \lambda_2 \|\mathcal{Y} - \mathcal{G} * \mathcal{X}\|_F^2 \\ & + \lambda_3 \sum_i \sum_j \left\| \mathcal{Q} * \mathcal{X}_i - \mathcal{Q} * \mathcal{X}_j \right\|_F^2 W_{ij} \\ \text{s.t.} \quad & \mathcal{P} * \mathcal{X} = \mathcal{P} * \mathcal{L} + \mathcal{E}, \quad \mathcal{P}' * \mathcal{P} = \mathcal{I} \\ & \mathcal{L} = \mathcal{M} \quad \mathcal{H} * \mathcal{P} = \mathcal{G}, \quad \mathcal{P} = \mathcal{Q} \\ & \sum_i \|\mathcal{Q} * \mathcal{X}_i\|_F^2 \ d_{ii} = 1. \end{aligned} \tag{18}$$

By introducing the Lagrange multipliers and penalty parameters for the equality constraints, the augmented Lagrangian function of (18) is formulated as follows:

$$\begin{aligned}& \mathcal{L}(\mathcal{M},\mathcal{L},\mathcal{E},\mathcal{H},\mathcal{G},\mathcal{P},\mathcal{Q}) \\ & = \|\mathcal{M}\|_* + \lambda_1 \|\mathcal{E}\|_1 + \lambda_2 \|\mathcal{Y} - \mathcal{G} * \mathcal{X}\|_F^2 \\ & + \lambda_3 \sum_i \sum_j \left\| \mathcal{Q} * \mathcal{X}_i - \mathcal{Q} * \mathcal{X}_j \right\|_F^2 W_{ij} \\ & + \frac{\rho}{2} \left(\left\| \mathcal{P} * \mathcal{X} - \mathcal{P} * \mathcal{L} - \mathcal{E} + \frac{\mathcal{C}_1}{\rho} \right\|_F^2 + \left\| \mathcal{L} - \mathcal{M} + \frac{\mathcal{C}_2}{\rho} \right\|_F^2 \right. \\ & \left. + \left\| \mathcal{H} * \mathcal{P} - \mathcal{G} + \frac{\mathcal{C}_3}{\rho} \right\|_F^2 + \left\| \mathcal{P} - \mathcal{Q} + \frac{\mathcal{C}_4}{\rho} \right\|_F^2 \right) \\ & \text{s.t.} \quad \sum_i \|\mathcal{Q} * \mathcal{X}_i\|_F^2 \ d_{ii} = 1 \end{aligned} \tag{19}$$

where $\mathcal{C}_1, \mathcal{C}_2, \mathcal{C}_3$, and $\mathcal{C}_4$ are the Lagrange multipliers and $\rho > 0$ is the penalty parameter. Equation (19) can be alternatively optimized with respect to seven subproblems as follows.

1) Updating $\mathcal{M}$: Fixing all variables except $\mathcal{M}$, we can restate the minimization of (19) as

$$\min_{\mathcal{M}} \frac{1}{\rho} \|\mathcal{M}\|_* + \frac{1}{2} \left\| \mathcal{M} - \left(\mathcal{L} + \frac{\mathcal{C}_2}{\rho} \right) \right\|_F^2. \tag{20}$$

As a TNN minimization problem [38], the closed-form solution to (20) can be derived using Theorem 1 below.

Theorem 1 [38]: Denote $\tau > 0$ $\|\cdot\|_*$ is nuclear-norm and $\mathcal{A}, \mathcal{B} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, the optimization problem stated as

$$\min_{\mathcal{A}} \tau \|\mathcal{A}\|_* + \frac{1}{2} \|\mathcal{A} - \mathcal{B}\|_F^2 \tag{21}$$

where $\|\cdot\|_*$ denotes the nuclear norm. The following tensor Singular Value Thresholding (t-SVT) method provides a global optimal solution to (21)

$$\mathcal{A} = \Omega_\tau(\mathcal{B}) = \mathcal{U} * \Omega_\tau(\mathcal{S}) * \mathcal{V}' \tag{22}$$

where $\mathcal{B} = \mathcal{U} * \mathcal{S} * \mathcal{V}'$ and $\Omega_\tau(\mathcal{S}) = \mathcal{S} * \mathcal{J}$. Here, $\mathcal{J}$ denotes an $n_1 \times n_2 \times n_3$ f-diagonal tensor, with its Fourier-domain diagonal element given by $\widetilde{\mathcal{J}}(i,i,j) = (1 - (\tau / \widetilde{\mathcal{S}}(i,i,j)))_+$.

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According to Theorem 1, the optimal solution $\mathcal{M}$ for the optimization problem in (20) is

$$\mathcal{M}_{\text{new}} = \Omega_{\frac{1}{\rho}}\left(\mathcal{L} + \frac{\mathcal{C}_2}{\rho}\right). \tag{23}$$

2) Updating $\mathcal{L}$: By holding all other variables constant except $\mathcal{L}$, we formulate the subproblem for solving $\mathcal{L}$ as

$$\min_{\mathcal{L}} \frac{\rho}{2}\left(\left\|\mathcal{P} * \mathcal{X} - \mathcal{P} * \mathcal{L} - \mathcal{E} + \frac{\mathcal{C}_1}{\rho}\right\|_F^2 + \left\|\mathcal{L} - \mathcal{M} + \frac{\mathcal{C}_2}{\rho}\right\|_F^2\right). \tag{24}$$

According to Property 1 in Section II-A, the above optimization problem can be transformed into the Fourier domain. Specifically, we perform the Fourier transform on the third-order tensors in the original domain as $\widetilde{\mathcal{X}} = \text{fft}(\mathcal{X})$. Then, we calculate the block-diagonal form of $\widetilde{\mathcal{X}}$ in the Fourier domain. The resulting equivalent formulation of (24) is

$$\min_{\overline{\mathbf{L}}} \left\|y\overline{\mathbf{P}}\overline{\mathbf{X}} - \overline{\mathbf{P}}\overline{\mathbf{L}} - \overline{\mathbf{E}} + \frac{\overline{\mathbf{C}}_1}{\mu}\right\|_F^2 + \left\|\overline{\mathbf{L}} - \overline{\mathbf{M}} + \frac{\overline{\mathbf{C}}_2}{\mu}\right\|_F^2 \tag{25}$$

where $\overline{\mathbf{X}}$, $\overline{\mathbf{P}}$, $\overline{\mathbf{L}}$, $\overline{\mathbf{E}}$, $\overline{\mathbf{M}}$, $\overline{\mathbf{C}}_1$, and $\overline{\mathbf{C}}_2$ are the block-diagonal forms of the tensor $\mathcal{X}$, $\mathcal{P}$, $\mathcal{L}$, $\mathcal{E}$, $\mathcal{M}$, $\mathcal{C}_1$, $\mathcal{C}_2$ in the Fourier domain, respectively.

For (25), we take the derivative of $\overline{\mathbf{L}}$ and setting the derivative to zero allows computing the optimal $\overline{\mathbf{L}}$ as

$$\overline{\mathbf{L}}_{\text{new}} = \left(\overline{\mathbf{P}}^T\overline{\mathbf{P}} + \overline{\mathbf{I}}\right)^{-1} \times \left(\overline{\mathbf{P}}^T\left(\overline{\mathbf{P}}\overline{\mathbf{X}} - \overline{\mathbf{E}} + \frac{\overline{\mathbf{C}}_1}{\mu}\right) + \left(\overline{\mathbf{M}} - \frac{\overline{\mathbf{C}}_2}{\mu}\right)\right). \tag{26}$$

We can obtain $\widetilde{\mathcal{L}}_{\text{new}}$ by restructuring $\overline{\mathbf{L}}_{\text{new}}$ into a third-order tensor, and then $\mathcal{L}_{\text{new}}$ can be derived through the inverse Fourier transform, expressed as $\mathcal{L}_{\text{new}} = \text{ifft}(\widetilde{\mathcal{L}}_{\text{new}})$.

3) Updating $\mathcal{E}$: With all other variables except $\mathcal{E}$ fixed, we can rewrite the minimization of (19) as

$$\min_{\mathcal{E}} \frac{\lambda_1}{\rho}\|\mathcal{E}\|_1 + \frac{1}{2}\left\|\mathcal{E} - \left(\mathcal{P} * \mathcal{X} - \mathcal{P} * \mathcal{L} + \frac{\mathcal{C}_1}{\rho}\right)\right\|_F^2. \tag{27}$$

The closed-form solution of (27) can be obtained by the following Theorem 2.

Theorem 2 [29]: Denote $\tau > 0$ and $\mathcal{A}, \mathcal{B} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, then the optimization problem

$$\min_{\mathcal{A}} \tau\|\mathcal{A}\|_1 + \frac{1}{2}\|\mathcal{A} - \mathcal{B}\|_F^2 \tag{28}$$

has a global optimal solution

$$\mathcal{A} = \mathfrak{S}_\tau(\mathcal{B}) \tag{29}$$

where $\mathfrak{S}_\tau(\cdot)$ represents the soft-thresholding operator, specified as

$$\mathfrak{S}_\tau(x) = \text{sgn}(x)\max(|x| - \tau, 0). \tag{30}$$

According to Theorem 2, the optimal solution $\mathcal{E}_{\text{new}}$ of (27) is

$$\mathcal{E}_{\text{new}} = \mathfrak{S}_{\frac{\lambda_1}{\rho}}\left(\mathcal{P} * \mathcal{X} - \mathcal{P} * \mathcal{L} + \frac{\mathcal{C}_1}{\rho}\right). \tag{31}$$

4) Updating $\mathcal{H}$: Keeping all variables except $\mathcal{H}$ unchanged in (19), we can rephrase the problem as follows:

$$\min_{\mathcal{H}} \frac{\rho}{2}\left\|\mathcal{H} * \mathcal{P} - \mathcal{G} + \frac{\mathcal{C}_3}{\rho}\right\|_F^2 \quad \text{s.t. } \mathcal{H}' * \mathcal{H} = \mathcal{I}. \tag{32}$$

Similar to *Updating* $\mathcal{L}$, we can solve this optimization problem using an equivalent Fourier-domain formulation, which is given by

$$\begin{aligned}\overline{\mathbf{H}}_{\text{new}} &= \arg\min \frac{\rho}{2}\left\|\overline{\mathbf{H}}\overline{\mathbf{P}} - \overline{\mathbf{G}} + \frac{\overline{\mathbf{C}}_3}{\rho}\right\|_F^2 \\ &= \arg\min \frac{\rho}{2}\left\|\overline{\mathbf{H}}\overline{\mathbf{P}} - \left(\overline{\mathbf{G}} - \frac{\overline{\mathbf{C}}_3}{\rho}\right)\right\|_F^2 \\ &\text{s.t. } \overline{\mathbf{H}}^T\overline{\mathbf{H}} = \overline{\mathbf{I}}.\end{aligned} \tag{33}$$

Theorem 3 [44]: For two given matrices A and B, the constrained optimization problem specified as

$$\begin{aligned}\mathbf{S} &= \arg\min \|\mathbf{S}\mathbf{A} - \mathbf{B}\|_F^2 \\ &\text{s.t. } \mathbf{S}^T\mathbf{S} = \mathbf{I}\end{aligned} \tag{34}$$

is corresponds to finding the orthogonal matrix $\mathbf{S}$ closest to the given matrix $\mathbf{B}\mathbf{A}^T$. Let the Singular-Value Decomposition (SVD) of $\mathbf{B}\mathbf{A}^T$ be $\mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$, then $\mathbf{S} = \mathbf{U}\mathbf{V}^T$.

Based on Theorem 3, the problem (33) can be readily solved via SVD, and the solution is given by

$$\text{SVD}\left(\mathbf{B}\mathbf{A}^T\right) = \text{SVD}\left(\left(\overline{\mathbf{G}} - \frac{\overline{\mathbf{C}}_1}{\rho}\right)\overline{\mathbf{P}}^T\right) = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T. \tag{35}$$

Thus, we can obtain the block-diagonal form in the Fourier domain as $\overline{\mathbf{H}}_{\text{new}} = \mathbf{U}\mathbf{V}^T$. Rewrite $\overline{\mathbf{H}}_{\text{new}}$ in the form of a third-order tensor $\widetilde{\mathcal{H}}_{\text{new}}$, and then $\mathcal{H}_{\text{new}} = \text{ifft}(\widetilde{\mathcal{H}}_{\text{new}})$.

5) Updating $\mathcal{G}$: With all variables except $\mathcal{G}$ already specified, the subproblem for calculating $\mathcal{G}$ is presented as

$$\min_{\mathcal{G}} \lambda_2\|\mathcal{Y} - \mathcal{G} * \mathcal{X}\|_F^2 + \frac{\rho}{2}\left\|\mathcal{H} * \mathcal{P} - \left(\mathcal{G} - \frac{\mathcal{C}_3}{\rho}\right)\right\|_F^2 \quad \text{s.t. } \mathcal{G}^T * \mathcal{G} = \mathcal{I}. \tag{36}$$

Analogously to *Updating* $\mathcal{H}$, we transform (36) to the Fourier domain

$$\min_{\overline{\mathbf{G}}} \lambda_2\left\|\overline{\mathbf{G}}\overline{\mathbf{X}} - \overline{\mathbf{Y}}\right\|_F^2 + \frac{\rho}{2}\left\|\overline{\mathbf{G}} - \left(\overline{\mathbf{H}}\overline{\mathbf{P}} + \frac{\overline{\mathbf{C}}_3}{\rho}\right)\right\|_F^2 \quad \text{s.t. } \overline{\mathbf{G}}^T\overline{\mathbf{G}} = \overline{\mathbf{I}}. \tag{37}$$

Both terms in (37) can also be solved using Theorem 3

$$\overline{\mathbf{G}}_{\text{new}} = \mathbf{U}\mathbf{V}^T \tag{38}$$

where $\mathbf{U}$ and $\mathbf{V}$ are the left-singular and right-singular vectors of $\lambda_2(\overline{\mathbf{Y}}\overline{\mathbf{X}}^T) + \rho/2(\overline{\mathbf{H}}\overline{\mathbf{P}} + (\overline{\mathbf{C}}_3)/\rho)$. After reorganizing $\overline{\mathbf{G}}_{\text{new}}$ into third-order tensor $\widetilde{\mathcal{G}}_{\text{new}}$, $\mathcal{G}_{\text{new}}$ can be recovered as $\mathcal{G}_{\text{new}} = \text{ifft}(\widetilde{\mathcal{G}}_{\text{new}})$.

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6) Updating $\mathcal{P}$: When all variables other than $\mathcal{P}$ are fixed, the minimization problem in (19) can be reformulated as

$$\min_{\mathcal{P}} \frac{\rho}{2}\left(\left\|\mathcal{P}*\mathcal{X}-\mathcal{P}*\mathcal{L}-\left(\mathcal{E}-\frac{\mathcal{C}_1}{\rho}\right)\right\|_F^2 + \left\|\mathcal{H}*\mathcal{P}-\left(\mathcal{G}-\frac{\mathcal{C}_3}{\rho}\right)\right\|_F^2 + \left\|\mathcal{P}-\mathcal{Q}+\frac{\mathcal{C}_4}{\rho}\right)\right\|_F^2$$
$$\text{s.t. } \mathcal{P}^T*\mathcal{P}=\mathcal{I}. \tag{39}$$

Mirroring the procedure of *Updating* $\mathcal{G}$, transform (39) into the Fourier domain:

$$\min_{\overline{\mathbf{P}}} \frac{\rho}{2}\left(\left\|\overline{\mathbf{P}}\left(\overline{\mathbf{X}}-\overline{\mathbf{L}}\right)-\left(\overline{\mathbf{E}}-\frac{\overline{\mathbf{C}}_1}{\rho}\right)\right\|_F^2 + \left\|\overline{\mathbf{H}}\overline{\mathbf{P}}-\left(\overline{\mathbf{G}}-\frac{\overline{\mathbf{C}}_3}{\rho}\right)\right\|_F^2 + \left\|\overline{\mathbf{P}}-\left(\overline{\mathbf{Q}}-\frac{\overline{\mathbf{C}}_4}{\rho}\right)\right\|_F^2\right)$$
$$\text{s.t. } \overline{\mathbf{P}}^T\overline{\mathbf{P}}=\overline{\mathbf{I}}. \tag{40}$$

All three terms of (40) are also solvable via Theorem 3, and the solution is as follows:

$$\text{SVD}\left(\left(\overline{\mathbf{E}}-\frac{\overline{\mathbf{C}}_1}{\rho}\right)\left(\overline{\mathbf{X}}-\overline{\mathbf{L}}\right)^T + \left(\overline{\mathbf{G}}-\frac{\overline{\mathbf{C}}_3}{\rho}\right)^T\overline{\mathbf{H}} + \left(\overline{\mathbf{Q}}-\frac{\overline{\mathbf{C}}_4}{\rho}\right)\right)=\mathbf{U}\Sigma\mathbf{V}^T. \tag{41}$$

Algorithm 1 Optimization of $\mathcal{P}$

Input: Third-order tensors $\mathcal{X},\mathcal{L},\mathcal{E},\mathcal{H},\mathcal{G},\mathcal{Q},\mathcal{C}_1,\mathcal{C}_3,\mathcal{C}_4$, and the parameter ρ;

1: Transform these third-order tensors into Fourier domain by $\widetilde{\mathcal{X}}$, $\widetilde{\mathcal{L}}$, $\widetilde{\mathcal{E}}$, $\widetilde{\mathcal{H}}$, $\widetilde{\mathcal{G}}$, $\widetilde{\mathcal{Q}}$, $\widetilde{\mathcal{C}}_1$, $\widetilde{\mathcal{C}}_3$, $\widetilde{\mathcal{C}}_4$ = fft($\mathcal{X}$, $\mathcal{L}$, $\mathcal{E}$, $\mathcal{H}$, $\mathcal{G}$, $\mathcal{Q}$, $\mathcal{C}_1$, $\mathcal{C}_3$, $\mathcal{C}_4$);

2: Compute block-diagonal matrix $\overline{\mathbf{X}}$, $\overline{\mathbf{L}}$, $\overline{\mathbf{E}}$, $\overline{\mathbf{H}}$, $\overline{\mathbf{G}}$, $\overline{\mathbf{Q}}$, $\overline{\mathbf{C}}_1$, $\overline{\mathbf{C}}_3$, and $\overline{\mathbf{C}}_4$ by Property 1;

3: Perform SVD on $\left(\overline{\mathbf{E}}-\frac{\overline{\mathbf{C}}_1}{\rho}\right)(\overline{\mathbf{X}}-\overline{\mathbf{L}})^T+\left(\overline{\mathbf{G}}-\frac{\overline{\mathbf{C}}_3}{\rho}\right)^T\overline{\mathbf{H}}+\left(\overline{\mathbf{Q}}-\frac{\overline{\mathbf{C}}_4}{\rho}\right)$ to get $\mathbf{U}$, $\mathbf{V}$ by Theorem 3;

4: Update $\overline{\mathbf{P}}$ by $\overline{\mathbf{P}}=\mathbf{U}\mathbf{V}^T$;

5: Reorganize $\overline{\mathbf{P}}$ into third-order tensor $\widetilde{\mathcal{P}}$;

6: Convert $\widetilde{\mathcal{P}}$ into original domain $\mathcal{P}$ by Inverse Fourier Transform ifft($\widetilde{\mathcal{P}}$);

Output: $\mathcal{P}$.

Thus, we have $\overline{\mathbf{P}}_{\text{new}}=\mathbf{U}\mathbf{V}^T$, and then operate $\overline{\mathbf{P}}_{\text{new}}$ into third-order tensor $\widetilde{\mathcal{P}}_{\text{new}}$. The optimal tensor $\mathcal{P}_{\text{new}}$ is obtained as ifft($\widetilde{\mathcal{P}}_{\text{new}}$). The procedure of optimizing $\mathcal{P}$ is completely summarized in Algorithm 1.

7) Updating $\mathcal{Q}$: Given every variable except $\mathcal{Q}$, the subproblem for finding $\mathcal{Q}$ is set up as follows:

$$\min_{\mathcal{Q}} \lambda_3\sum_{i,j}\left\|\mathcal{Q}*\mathcal{X}_i-\mathcal{Q}*\mathcal{X}_j\right\|_F^2 W_{ij}+\frac{\rho}{2}\left\|\mathcal{P}-\mathcal{Q}+\frac{\mathcal{C}_4}{\rho}\right\|_F^2$$
$$\text{s.t. } \sum_i\|\mathcal{Q}*\mathcal{X}_i\|_F^2\ d_{ii}=1. \tag{42}$$

Convert (42) to the Fourier domain

$$\min_{\mathcal{Q}} \lambda_3\sum_{i,j}\left\|\overline{\mathbf{Q}}\,\overline{\mathbf{X}}_i-\overline{\mathbf{Q}}\,\overline{\mathbf{X}}_j\right\|_F^2 W_{ij}+\frac{\rho}{2}\left\|\overline{\mathbf{P}}-\overline{\mathbf{Q}}+\frac{\overline{\mathbf{C}}_4}{\rho}\right\|_F^2$$
$$\text{s.t. } \sum_i\left\|\overline{\mathbf{Q}}\,\overline{\mathbf{X}}_i\right\|_F^2\ d_{ii}=1. \tag{43}$$

Since $\|A\|_F^2=\text{tr}(AA^T)$, the following equivalent derivations can be obtained:

$$\begin{aligned}&\sum_{i,j}\left\|\overline{\mathbf{Q}}*\overline{\mathbf{X}}_i-\overline{\mathbf{Q}}*\overline{\mathbf{X}}_j\right\|_F^2 W_{ij}\\ &=\sum_{i,j}\text{tr}\left(\overline{\mathbf{Q}}\left(\left(\overline{\mathbf{X}}_i-\overline{\mathbf{X}}_j\right)\left(\overline{\mathbf{X}}_i-\overline{\mathbf{X}}_j\right)^T W_{ij}\right)\overline{\mathbf{Q}}^T\right)\\ &=\text{tr}\left(\overline{\mathbf{Q}}\left(\sum_{ij}\left(\overline{\mathbf{X}}_i-\overline{\mathbf{X}}_j\right)\left(\overline{\mathbf{X}}_i-\overline{\mathbf{X}}_j\right)^T W_{ij}\right)\overline{\mathbf{Q}}^T\right)\\ &=\text{tr}\left(\overline{\mathbf{Q}}\mathbf{L}\overline{\mathbf{Q}}^T\right)\end{aligned} \tag{44}$$

where $\mathbf{L}=\sum_{i,j}(\overline{\mathbf{X}}_i-\overline{\mathbf{X}}_j)(\overline{\mathbf{X}}_i-\overline{\mathbf{X}}_j)^T W_{ij}$. Similarly, we can obtain

$$\begin{aligned}&\sum_i\left\|\overline{\mathbf{Q}}\,\overline{\mathbf{X}}_i\right\|_F^2\ d_{ii}\\ &=\sum_i\text{tr}\left(\overline{\mathbf{Q}}\left(\overline{\mathbf{X}}_i\overline{\mathbf{X}}_i^T d_{ii}\right)\overline{\mathbf{Q}}^T\right)\\ &=\text{tr}\left(\overline{\mathbf{Q}}\left(\sum_i\left(\overline{\mathbf{X}}_i\overline{\mathbf{X}}_i^T d_{ii}\right)\right)\overline{\mathbf{Q}}^T\right)\\ &=\text{tr}\left(\overline{\mathbf{Q}}\mathbf{D}\overline{\mathbf{Q}}^T\right)\end{aligned} \tag{45}$$

where $\mathbf{D}=\sum_i\overline{\mathbf{X}}_i\overline{\mathbf{X}}_i^T d_{ii}$. Therefore, the optimization problem (43) can be transformed into the following minimization problem of matrix-trace:

$$\min_{\overline{\mathbf{Q}}} \lambda_3\,\text{tr}\left(\overline{\mathbf{Q}}\mathbf{L}\overline{\mathbf{Q}}^T\right)+\frac{\rho}{2}\left\|\overline{\mathbf{P}}-\left(\overline{\mathbf{Q}}-\frac{\overline{\mathbf{C}}_4}{\rho}\right)\right\|_F^2$$
$$\text{s.t. } \text{tr}\left(\overline{\mathbf{Q}}\mathbf{D}\overline{\mathbf{Q}}^T\right)=1. \tag{46}$$

Adopt the gradient descent method based on (46). Introducing Lagrange multiplier β and constructing the Lagrange function

$$L\left(\overline{\mathbf{Q}},\beta\right)=\lambda_3\,\text{tr}\left(\overline{\mathbf{Q}}\mathbf{L}\overline{\mathbf{Q}}^T\right)+\frac{\rho}{2}\left\|\overline{\mathbf{P}}-\overline{\mathbf{Q}}+\frac{\overline{\mathbf{C}}_4}{\rho}\right\|_F^2+\beta\left(\text{tr}\left(\overline{\mathbf{Q}}\mathbf{D}\overline{\mathbf{Q}}^T\right)-1\right). \tag{47}$$

Find the gradients of (47) with respect to $\overline{\mathbf{Q}}$ and β, respectively,

$$\begin{aligned}\nabla\overline{\mathbf{Q}}&=2\lambda_3\overline{\mathbf{Q}}\mathbf{L}-\rho\left(\overline{\mathbf{P}}-\overline{\mathbf{Q}}+\frac{\overline{\mathbf{C}}_4}{\rho}\right)+2\beta\overline{\mathbf{Q}}\mathbf{D}\\ \nabla\beta&=\text{tr}\left(\overline{\mathbf{Q}}\mathbf{D}\overline{\mathbf{Q}}^T\right)-1.\end{aligned} \tag{48}$$

In the gradient-descent process, continuously iterate and update $\overline{\mathbf{Q}}_{\text{new}}$ and β_{new} according to the update formula $\overline{\mathbf{Q}}_{\text{new}}=\overline{\mathbf{Q}}-\gamma\nabla\overline{\mathbf{Q}}$ and $\beta_{\text{new}}=\beta-\gamma\nabla\beta$ (the learning rate γ) until the convergence condition is met.

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8) Updating Parameter $\mathcal{C}_1$, $\mathcal{C}_2$, $\mathcal{C}_3$, $\mathcal{C}_4$, and ρ: The Lagrangian multipliers $\mathcal{C}_1$, $\mathcal{C}_2$, $\mathcal{C}_3$, $\mathcal{C}_4$ and penalty parameter ρ are updated as follows:

$$\begin{aligned}\mathcal{C}_1 &= \mathcal{C}_1 + \rho(\mathcal{P} * \mathcal{X} - \mathcal{P} * \mathcal{L} - \mathcal{E}) \\ \mathcal{C}_2 &= \mathcal{C}_2 + \rho(\mathcal{L} - \mathcal{M}) \\ \mathcal{C}_3 &= \mathcal{C}_3 + \rho(\mathcal{H} * \mathcal{P} - \mathcal{G}) \\ \mathcal{C}_4 &= \mathcal{C}_4 + \rho(\mathcal{P} - \mathcal{Q}) \\ \rho &= \min\{\mu\rho, \rho_{\max}\} \end{aligned} \tag{49}$$

where ρ and $\rho_{\max}$ are constants, and μ is empirically set to 1.1 throughout this article.

Algorithm 2 Algorithm for Solving t-LRDGE Model in (17)

Input: Data tensor set $\mathcal{X} \in \mathbb{R}^{d\times N\times w^2}$, label tensor $\mathcal{Y} \in \mathbb{R}^{c\times r\times w^2}$, and parameters $\lambda_1, \lambda_2, \lambda_3, \rho, \rho_{\max}, \mu$

1: **Initialize:** Initialize W using Eq.(11); Initialize $\mathcal{P}$, $\mathcal{L}$, $\mathcal{E}$, $\mathcal{H}$, $\mathcal{M}$, $\mathcal{C}_1$, $\mathcal{C}_2$, $\mathcal{C}_3$ and $\mathcal{C}_4$ to zero tensors; $\rho = 0.1$, $\mu = 1.1$, $\rho_{\max} = 10^8$, and $\epsilon = 10^{-6}$.
2: **repeat**
3: Update $\mathcal{M}$ by Eq.(23);
4: Update $\mathcal{L}$ using Eq.(26);
5: Update $\mathcal{E}$ by solving Eq.(31);
6: Update $\mathcal{H}$ according to Theorem 3;
7: Update $\mathcal{G}$ using Eq.(38);
8: Update $\mathcal{P}$ by the Algorithm 1;
9: Update $\mathcal{Q}$ by Eq.(48);
10: Update parameters using Eq.(49);
11: **until** Convergence: $\|\mathcal{P}*\mathcal{X}-\mathcal{P}*\mathcal{L}-\mathcal{E}\|_F^2 \le \epsilon$, $\|\mathcal{L}-\mathcal{M}\|_F^2 \le \epsilon$, $\|\mathcal{H} * \mathcal{P} - \mathcal{G}\|_F^2 \le \epsilon$, $\|\mathcal{P} - \mathcal{Q}\|_F^2 \le \epsilon$

Output: $\mathcal{P}$, $\mathcal{L}$

The complete optimization procedure for solving the t-LRDGE model is summarized in Algorithm 2. When the residual is sufficiently small, the iterative optimization algorithm will converge and stop. Ultimately, the algorithm outputs the optimal projection tensor $\mathcal{P}$ and the denoised high-dimensional data $\mathcal{L}$. The low-dimensional representation $\mathcal{P}*\mathcal{L}$ is obtained via the t-product operation, achieving effective DR for hyperspectral data.

D. Complexity Analysis

The computational complexity of the proposed t-LRDGE model comprises two main components: offline graph construction and online iterative optimization. Specifically, the graph construction phase involves calculating covariance descriptors alongside the EMD matrix. Given N training samples with d spectral bands and a spatial mode size of n_3 (corresponding to w^2), this step is dominated by $\mathcal{O}(N^2 C_{\text{EMD}} + Nn_3d^2)$, where C_{EMD} denotes the cost of a single EMD calculation. This precomputation remains highly efficient because our method utilizes only a small subset of training samples instead of the entire image. Regarding the ADMM optimization, all operations are executed in the Fourier domain by leveraging t-product properties. For a tensor of size $d \times N \times n_3$, the FFT and IFFT operations incur a computational cost of $\mathcal{O}(dNn_3 \log n_3)$. In each iteration, the updates for the low-rank tensor $\mathcal{L}$ and the projection tensor $\mathcal{P}$ are computed slice by slice in the frequency domain. These update steps carry complexities of $\mathcal{O}(n_3d^2N)$ and approximately $\mathcal{O}(n_3(d^2N + d^3))$, respectively. Consequently, the overall computational complexity of t-LRDGE is bounded by approximately $\mathcal{O}(Kn_3d^2N)$, where K represents the total number of iterations. This linear dependence on both the spatial mode size n_3 and the sample count N guarantees that t-LRDGE is computationally efficient for high-dimensional HSI FE tasks.

IV. Experiments

A. Experimental Settings

1) Datasets Description: In this section, the performance of the proposed t-LRDGE model is verified via a series of experiments implemented on three representative HSI datasets: Indian Pines, Salinas, and WHU-Hi-LongKou. These training ratios were selected to be consistent with the experimental settings of the comparison methods to ensure fair comparison.

1) *Indian Pines:*: Indian Pines dataset, captured by the AVIRIS sensor in 1992 over an agricultural region in Northern Indiana, serves as the basis for our study. Originally covering a spectral range of 0.4–2.45 μm with 220 bands, the data were preprocessed to exclude 20 bands affected by water absorption (specifically bands 104–108, 150–163, and 220). Consequently, 200 spectral bands were retained for analysis. The image consists of 145 × 145 pixels and contains 16 distinct ground-truth classes. For the training phase, we randomly selected 10% of the available samples.
2) *Salinas:* The Salinas dataset, acquired over Southern California in 1998 utilizing the AVIRIS sensor, is characterized by a high spatial resolution of 3.7 m. The image dimensions are 512 × 217 pixels. In terms of spectral resolution, 204 bands were retained for analysis after discarding 20 bands prone to water absorption. The ground truth comprises 16 distinct classes. For our experimental setup, we randomly selected 1% of the labeled pixels for training, reserving the remaining samples for testing.
3) *WHU-Hi-LongKou [45]:* The Longkou dataset was acquired in 2018 over Hubei Province, China, utilizing a Headwall Nano-Hyperspec sensor mounted on an uncrewed aerial vehicle (UAV). This image features a spatial resolution of 550 × 400 pixels and consists of 270 spectral bands ranging from 400 to 1000 nm. The dataset includes nine distinct classes. For the training set, we utilized a very small proportion (0.2%) of the labeled samples.

2) Competitors: The classic supervised linear DR method, LDA, and the matrix-based embedding method LPP are utilized as baselines. TLPP, MTLPP, and WMTLPP are improved GE methods based on tensor theory. MNNCA and LSaCGE are collaborative representation (CR)-based GE methods. LSaCGE strives to preserve both local structure and collaborative relationships in the embedding process, while MNNCA focuses on incorporating multineighborhood and nonlinear

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constraints to adapt to HSI. On the other hand, the data recovery method TRPCA, based on low-rank decomposition, is also introduced as a baseline for evaluating robustness. tLTSL combines TRPCA with projection learning to realize the integrated operation of DR and robustness enhancement. L_2, p-RER focuses on employing the L_2, p-norm to achieve robust feature DR for HSI via low-rank decomposition. Similarly, LPSPER integrates LRR with projection learning and embedding regression to simultaneously perform FE, thereby enhancing overall robustness and improving feature discriminability. Differently, SSMM proposes a manifold learning framework that utilizes a spatial–spectral self-training module to augment label information and imposes regularized metric constraints on the manifold to ensure intraclass compactness and interclass separability. In the experiments, the parameters of the compared methods are set based on relevant references or grid search, and each experiment is repeated five times to reduce randomness.

3) Parameters: The proposed t-LRDGE model involves three tradeoff parameters: λ_1, λ_2, and λ_3. To determine their optimal values, we employed a grid search strategy over the candidate set $\{0.01, 0.1, 1, 10, 100\}$. The final parameter configurations selected for each of the three datasets are provided in Section IV-C2.

The spatial size of tensor samples plays a critical role in tensor-based methods. To determine the optimal setting, we evaluated window sizes ranging from $\{3\times3, 5\times5, \ldots, 11\times11\}$ across the three datasets. Section IV-C3 analyses the impact of different spatial window sizes on overall accuracy (OA). The results clearly indicate that performance varies with window size, maximizing at specific window sizes. Notably, a window size of 9×9 consistently yields the best results for Indian Pines, Salinas, and LongKou.

4) Classification Configurations: To quantitatively assess the performance of the features extracted via DR, we employed four evaluation indicators: OA, average accuracy (AA), per-class accuracy, and the Kappa coefficient (Kappa). The kNNs algorithm served as the classifier for all experiments. To ensure statistical stability and reduce sample bias, each experiment is repeated five times independently.

B. Experimental Results

In this section, we conduct experimental comparisons between the t-LRDGE model and its competitors. Overall, the t-LRDGE model demonstrates significant performance advantages in processing HSIs. The detailed experimental results are described below.

To evaluate the FE and classification ability of the t-LRDGE model in HSI, we selected Indian Pines, Salinas, and LongKou datasets. Quantitative comparisons with other different DR methods are detailed in Tables I–III, using OA, AA, and Kappa as evaluation indices. In addition, the corresponding classification maps are visually displayed in Figs. 3–5.

From the first four columns in Tables I–III, LPP, a classic GE method based on manifold learning, can preserve local geometric structures to a certain extent. TLPP, MTLPP, and WMTLPP are all variants of LPP and belong to the family of tensor-based methods. Apparently, the data in these

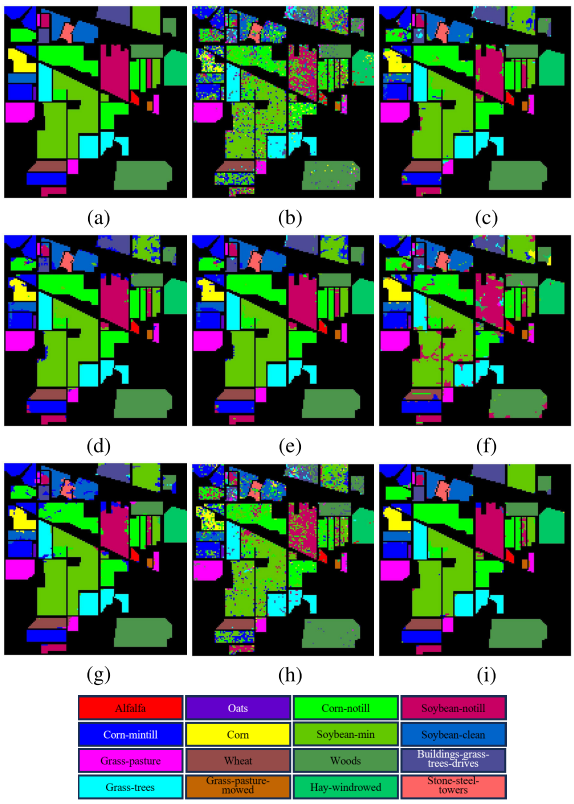


Fig. 3. kNN classification maps for Indian Pines using feature representation by different DR methods. (a) Ground truth. (b) LPP. (c) TLPP. (d) MTLPP. (e) WMTLPP. (f) TRPCA. (g) tLTSL. (h) MNNCA. (i) Ours.

tables intuitively demonstrate that tensor-based methods (e.g., TLPP, MTLPP, and WMTLPP) achieve significantly better performance compared with vector-based methods (e.g., LPP). Specifically, TLPP, MTLPP, and WMTLPP achieve performance improvements of 27.09%, 27.69%, and 27.27% in comparison with LPP, respectively, on Indian dataset; similar enhancements of 14.02%, 14.92%, and 14.65% are observed on the Salinas dataset; and the LongKou dataset also exhibits increases of 7.98%, 8.15%, and 8.39%. As clearly illustrated in Figs. 3–5, in the classification maps of matrix-based methods [Figs. 3–5(b) and (h)], misclassified points are distributed in a disorderly manner. In contrast, in the classification maps of tensor-based methods [Figs. 3–5(c)–(i)], misclassified points are mainly concentrated at class boundaries, with superior classification performance within classes. This is because tensor-based methods can fully explore and utilize the spatial–spectral information of HSI, thereby preserving its intrinsic information more efficiently.

Furthermore, both TRPCA and tLTSL decompose tensor data into a low-rank data component and a sparse part from a global perspective. As indicated by the results in three tables, although low-rank decomposition-based DR methods (e.g., TRPCA, tLTSL) are inferior in accuracy to tensor GE methods (e.g., TLPP, MTLPP, and WMTLPP) that consider local information, they exhibit significant advantages in terms of anti-noise performance. Relevant details will be elaborated in Section IV-C1. Our method not only outperforms them in noise resistance but also excels in classification accuracy

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TABLE I

CLASSIFICATION ACCURACY (%) FOR INDIAN PINES DATASET WITH KNN CLASSIFIER

Class	LDA [10]	LPP [16]	TLPP [23]	MTLPP [24]	WMTLPP [25]	TRPCA [29]	tLTSL [32]	MNNCA [46]	SSMM [47]	Ours
1	14.63	36.83	91.46	90.73	92.11	88.54	76.83	83.33	**95.12**	94.74
2	69.96	53.19	93.00	95.91	96.51	81.25	92.78	77.35	95.33	**98.68**
3	44.58	52.90	93.17	95.26	96.34	75.50	89.96	69.50	93.17	**97.15**
4	23.00	36.29	91.50	**94.23**	82.52	70.89	93.24	40.67	96.24	94.17
5	90.80	82.05	93.10	94.23	**99.77**	90.67	93.84	90.02	99.77	98.16
6	95.13	91.39	98.69	98.61	**99.85**	98.49	97.25	98.17	99.24	99.38
7	82.61	86.00	97.20	97.60	100.0	89.20	79.20	94.44	68.00	**100.0**
8	99.07	93.98	99.74	99.77	100.0	98.58	97.35	**100.0**	100.0	99.77
9	20.00	30.56	77.78	90.00	100.0	92.22	77.22	90.00	100.0	**100.0**
10	61.60	63.15	93.53	**96.07**	95.53	87.31	93.27	74.21	95.77	94.19
11	70.85	70.71	96.55	**98.26**	92.54	90.69	96.01	78.15	94.88	96.90
12	56.74	40.82	86.72	92.62	85.88	74.94	84.27	76.39	89.68	**96.42**
13	94.02	88.21	97.83	98.10	99.46	99.02	98.37	**100.0**	99.46	98.37
14	96.66	89.68	98.15	**99.07**	97.72	96.04	97.21	89.86	87.26	98.34
15	36.31	33.72	94.35	96.27	93.68	82.25	85.42	65.55	93.08	**99.14**
16	79.76	85.71	94.52	96.91	98.77	95.71	75.36	91.14	100.0	**98.77**
OA	71.91	67.92	95.01	95.61	95.19	88.16	93.55	80.38	94.47	**97.50**
AA	64.73	64.70	93.81	96.48	95.67	88.39	89.22	80.20	94.19	**97.76**
Kappa	67.74	63.34	94.31	96.91	94.52	86.50	93.53	78.00	93.70	**97.15**

TABLE II

CLASSIFICATION ACCURACY (%) FOR THE SALINAS DATASET WITH KNN CLASSIFIER

Class	LDA [10]	LPP [16]	TLPP [23]	MTLPP [24]	WMTLPP [25]	TRPCA [29]	tLTSL [32]	MNNCA [46]	L_2,p-RER [31]	LPSPER [48]	Ours
1	96.93	98.84	99.70	99.70	99.95	97.34	78.40	99.48	99.29	97.12	**100.0**
2	99.78	97.75	99.97	**100.0**	99.92	97.75	94.88	99.88	99.84	99.73	99.40
3	96.98	91.98	99.39	100.0	99.13	98.77	96.88	96.08	99.31	96.74	**100.0**
4	96.01	98.70	88.56	92.90	85.59	99.06	81.17	99.13	99.13	94.47	**99.20**
5	97.74	96.61	99.17	**99.66**	98.68	94.57	82.81	98.62	98.52	95.97	98.53
6	99.36	99.95	100.0	100.0	100.0	99.29	96.10	99.81	99.76	99.67	**100.0**
7	99.66	99.63	99.92	**100.0**	99.97	98.98	96.28	99.63	99.86	99.84	99.94
8	74.86	55.54	92.52	92.54	92.70	74.22	95.84	73.87	85.13	88.33	**95.99**
9	99.76	98.73	100.0	100.0	99.98	98.71	99.43	99.88	99.50	99.39	**100.0**
10	90.91	85.52	94.73	94.89	94.61	82.84	93.59	95.18	93.01	94.71	**97.47**
11	91.96	94.80	99.43	**100.0**	98.96	95.65	92.82	93.35	94.32	95.26	98.49
12	96.65	95.55	100.0	100.0	100.0	99.90	95.13	99.73	99.63	96.34	**100.0**
13	98.13	95.15	99.12	**100.0**	98.35	93.61	90.96	98.11	98.18	95.98	98.02
14	92.07	77.08	98.77	99.34	97.45	81.42	89.43	95.94	92.95	96.34	**99.34**
15	47.71	54.11	87.24	92.32	92.86	58.37	93.95	65.06	63.48	86.03	**98.03**
16	98.10	97.60	99.83	99.83	**99.94**	89.27	93.63	98.60	98.48	97.91	97.93
OA	86.11	81.96	95.98	96.88	96.61	86.22	93.85	88.71	90.99	95.85	**98.46**
AA	92.29	89.85	97.40	98.20	97.38	91.23	91.96	94.02	95.02	94.24	**98.90**
Kappa	84.52	79.97	95.52	96.52	96.23	84.65	93.15	87.45	98.29	93.59	**98.29**

TABLE III

CLASSIFICATION ACCURACY (%) FOR THE LONGKOU DATASET WITH KNN CLASSIFIER

Class	LDA [10]	LPP [16]	TLPP [23]	MTLPP [24]	WMTLPP [25]	TRPCA [29]	tLTSL [32]	MNNCA [46]	LSaCGE [49]	LPSPER [48]	Ours
1	94.49	91.62	99.71	99.71	**99.81**	96.72	98.75	98.84	93.46	98.94	99.72
2	37.23	52.36	89.33	81.48	87.17	75.03	78.64	67.50	55.55	62.71	**96.04**
3	13.55	58.48	80.33	83.01	88.23	47.50	79.17	88.35	49.99	89.47	**97.75**
4	93.05	90.50	97.86	**99.21**	99.02	95.82	97.12	97.43	90.58	93.10	98.92
5	12.94	54.57	88.87	**92.18**	90.49	63.41	83.61	59.74	83.20	71.00	88.00
6	96.29	98.48	99.04	99.27	98.26	89.16	96.39	98.17	**99.64**	90.52	98.91
7	99.93	99.98	99.99	99.98	99.99	99.95	99.96	99.91	99.97	99.84	**99.99**
8	42.69	74.78	81.65	82.43	82.66	62.42	70.81	86.84	85.39	**89.30**	80.18
9	36.19	29.74	**89.60**	86.63	88.93	34.74	50.04	72.79	53.68	64.79	79.17
OA	87.44	89.39	97.37	97.54	97.78	91.99	94.87	95.40	91.16	84.35	**98.43**
AA	58.48	72.28	91.82	91.54	92.73	73.86	83.83	85.51	79.02	93.46	**93.19**
Kappa	83.16	85.98	96.55	96.75	89.37	84.95	93.23	88.01	88.00	91.42	**97.14**

on the three datasets. Specifically, it surpasses TRPCA by 9.34%, 12.24%, and 6.44% on Indian Pines, Salinas, and LongKou datasets, respectively, and outperforms tLTSL by 3.95%, 4.61%, and 3.56% on the same three datasets, respectively. $L_{2,p}$-RER adopts the $L_{2,p}$-norm for optimization in the process of low-rank decomposition. Our method is 7.47% higher than $L_{2,p}$-RER on the Salinas dataset. Comparisons were also conducted with the manifold learning method SSMM and the low-rank selective projection method LPSPER. SSMM achieves an OA of 94.47% on Indian Pines dataset.

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TABLE IV
CLASSIFICATION ACCURACY (%) OF KNN CLASSIFIER FOR THREE DATASETS WITH NOISE

Dataset		METHODS							
		LPP	TLPP	MTLPP	WMTLPP	TRPCA	tLTSL	MNNCA	Ours
Indian Pines	OA	55.56	87.32	86.26	87.55	80.73	89.95	69.67	**96.73**
	AA	51.64	90.65	88.35	88.64	78.79	92.58	69.76	**95.63**
	Kappa	49.02	85.57	84.98	85.79	77.87	89.92	65.20	**95.12**
Salinas	OA	70.90	86.49	90.41	88.74	84.81	88.15	82.89	**94.26**
	AA	72.51	91.08	93.24	89.93	90.22	89.57	87.32	**95.86**
	Kappa	67.63	84.82	91.11	88.61	83.10	85.64	80.93	**93.61**
LongKou	OA	79.98	72.94	89.82	91.02	86.98	89.23	88.95	**93.85**
	AA	53.46	49.32	72.94	72.59	69.99	73.94	62.53	**80.80**
	Kappa	73.56	68.81	82.10	83.36	85.47	86.54	85.16	**91.83**

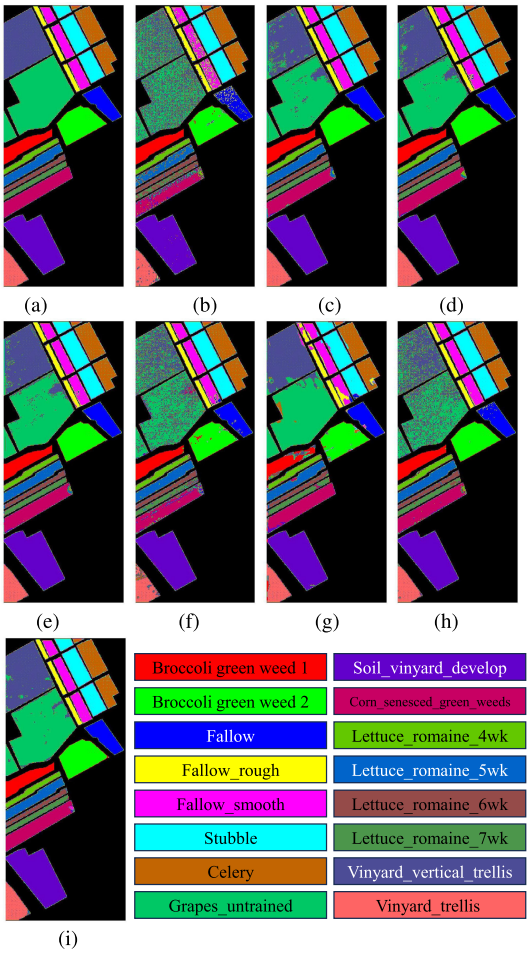


Fig. 4. KNN Classification maps of different methods on the Salinas dataset. (a) Ground truth. (b) LPP. (c) TLPP. (d) MTLPP. (e) WMTLPP. (f) TRPCA. (g) tLTSL. (h) MNNCA. (i) Ours.

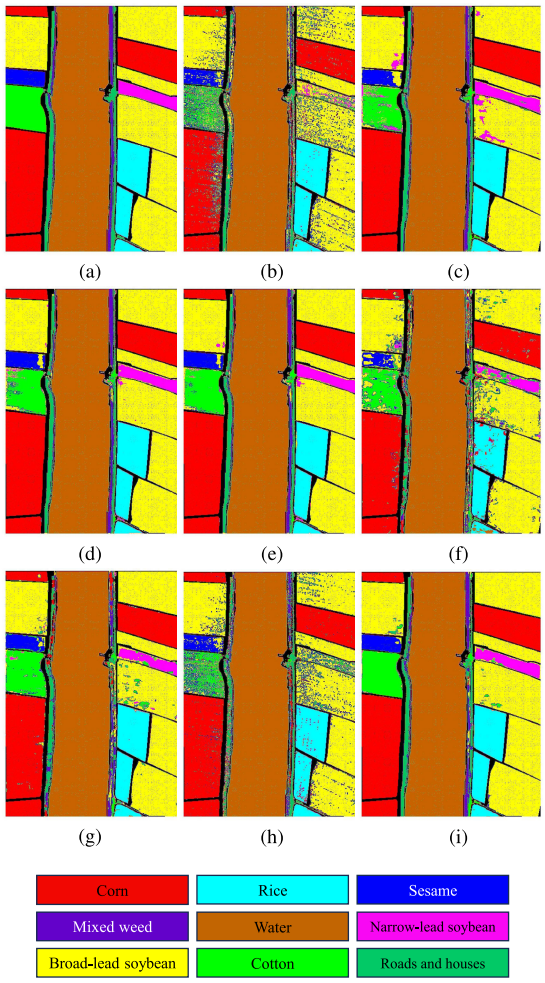


Fig. 5. KNN Classification maps of different methods on the LongKou dataset. (a) Ground truth. (b) LPP. (c) TLPP. (d) MTLPP. (e) WMTLPP. (f) TRPCA. (g) tLTSL. (h) MNNCA. (i) Ours.

Meanwhile, LPSPER attains OAs of 95.85% and 84.35% on the Salinas and LongKou datasets, respectively. Our proposed method consistently outperforms these advanced techniques, exceeding SSMM by 3.03% on Indian Pines, and surpassing LPSPER by 2.61% and 14.08% on the Salinas and LongKou datasets, respectively.

Finally, MNNCA, as a CR-based GE method, achieves dual-level neighborhood reconstruction to search for the optimal neighborhood and explore the common attributes within the neighborhood. It can be seen from Tables I–III that the OA of MNNCA reaches 80.38%, 88.71%, and 95.40% on Indian Pines, Salinas, and LongKou datasets, respectively. LSaCGE is also a CR-based GE method, and its OA reaches 91.16% on the LongKou dataset.

C. Model Analysis

1) Robustness Analysis: To validate the robustness and performance advantages of the t-LRDGE method in complex noise environments, we constructed a composite noise model

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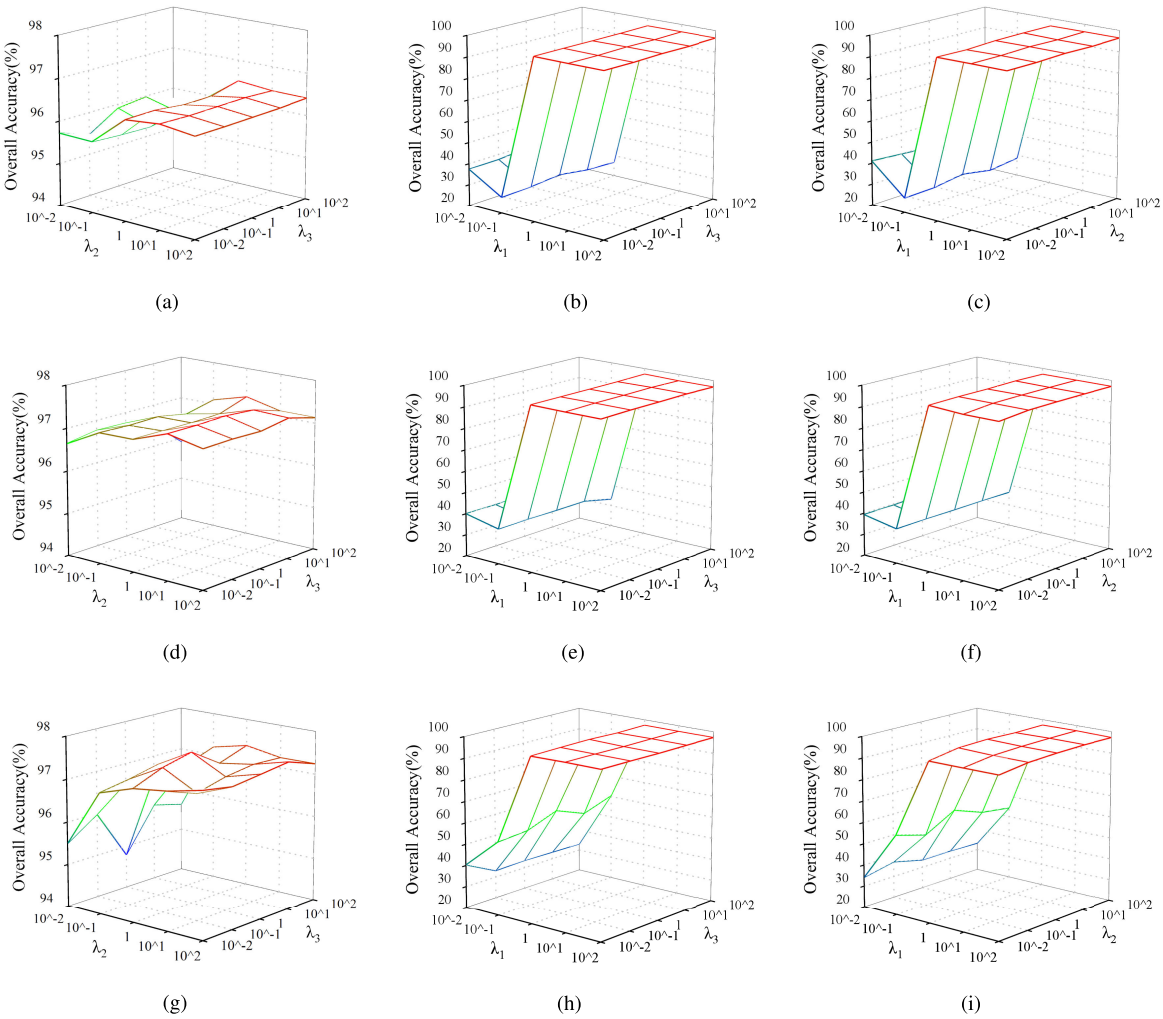


Fig. 6. Performance of t-LRDGE model over different parameter settings. (a)–(c) Correspond to results for Indian Pines dataset with $\lambda_1 = 1$, $\lambda_2 = 1$, and $\lambda_3 = 1$, respectively. (d)–(f) Correspond to results for the Salinas dataset with $\lambda_1 = 1$, $\lambda_2 = 1$, and $\lambda_3 = 1$, respectively. (g)–(i) Correspond to results for the LongKou dataset with $\lambda_1 = 1$, $\lambda_2 = 1$, and $\lambda_3 = 1$, respectively.

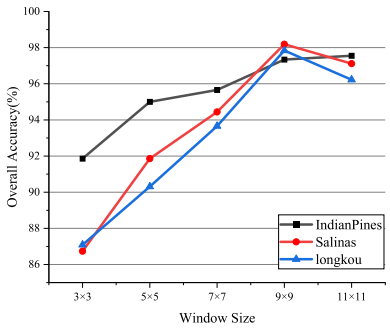


Fig. 7. Classification accuracy with different spatial window sizes.

that introduces two types of interference. Specifically, zero-mean additive Gaussian noise is added to random spectral bands with variances distributed between 0 and 0.5, and salt-and-pepper noise is applied to a subset of 30 randomly chosen bands with noise density ratios varying from 0 to 0.3.

The specific classification results are presented in Table IV on Indian Pines, Salinas, and LongKou datasets. As can be seen from this table, under the interference of composite noise, the performance of all methods degrades to varying degrees. Among GE methods, LPP, TLPP, MTLPP, and WMTLPP exhibit accuracy decreases of 12.36%, 11.06%, 9.41%; 7.69%, 9.49%, 24.43%; 9.35%, 6.47%, 7.72%; and 7.64%, 7.87%, 6.76% across the three datasets, respectively. In contrast, low-rank decomposition-based methods such as TRPCA, tLTSL, and t-LRDGE (Ours) show significantly smaller reductions: 7.43%, 1.41%, 5.01% for TRPCA; 3.60%, 5.70%, 5.64% for tLTSL; and 0.77%, 4.2%, 3.98% for t-LRDGE across the three datasets. These results clearly indicate that when facing such composite noise, the low-rank decomposition-based methods exhibit significantly better anti-noise performance than the GE methods. It is because GE methods rely on local structures to construct graphs, and noise easily causes deviations in graph construction, which in turn directly affects classification accuracy. In contrast, low-rank decomposition methods can effectively preserve the essential features of data by separating

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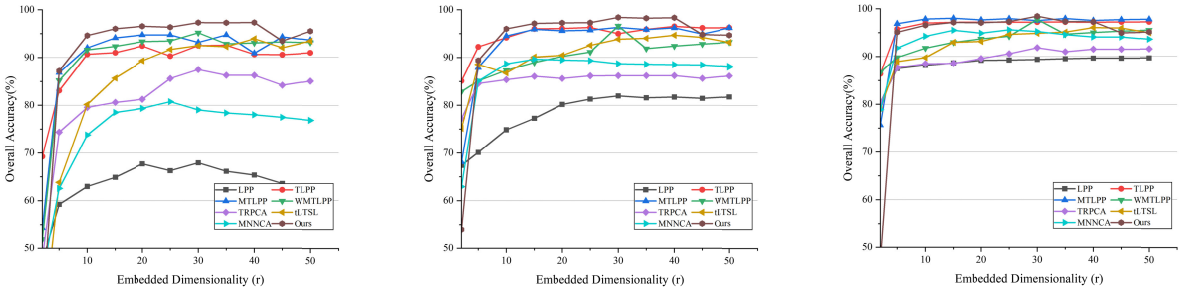


Fig. 8. Classification accuracy with different embedded dimensionality r. (a) Indian Pines. (b) Salinas. (c) LongKou.

the sparse residual part, thus having stronger anti-interference capability. These experimental results quantitatively demonstrate that t-LRDGE can effectively suppress noise interference and maintain stable performance, validating the robustness of the proposed model.

2) Parameters Sensitivity Analysis: There are three tradeoff parameters in (17), where λ_1 is dedicated to the global-preserving low-rank module, λ_2 is used to balance the LGE module, and λ_3 serves to balance the DE module. The three tradeoff parameters λ_1, λ_2, and λ_3 are tuned within the range of {0.01, 0.1, 1, 10, 100}. By fixing one parameter to a specific value and analyzing the impact of the remaining two, we are able to thoroughly assess the performance of the t-LRDGE model under varying parameter configurations.

As illustrated in Fig. 6, we notice that λ_1 has a significant impact on the final accuracy of the DR model: when $\lambda_1 < 1$, the accuracy remains at a low level; when $\lambda_1 \geq 1$, the accuracy shows a significant improvement. This is because the final classification accuracy of the t-LRDGE model is computed from the low-dimensional denoised data generated by the GLR module, whose performance directly dictates the efficacy of subsequent classification. As a key adjustment parameter of this module, the rationality of λ_1 plays a decisive role in the final result. Therefore, when applying to a new database, λ_1 should be regarded as the primary parameter for optimization. Ultimately, the experimental results indicate that the optimal settings for $\{\lambda_1, \lambda_2, \lambda_3\}$ are {1, 10, 0.1} for Indian Pines, {1, 0.01, 1} for Salinas, and {1, 1, 1} for LongKou.

3) Window Size Analysis: The spatial size of tensor samples, denoted as $w \times w$, is a critical parameter in our proposed t-LRDGE model. It determines the extent of local spatial neighborhood information incorporated into the tensor representation. A window that is too small may fail to capture sufficient spatial structural consistency and texture information, making the model sensitive to noise. Conversely, an excessively large window may introduce interference from pixels of different classes, thereby degrading discriminative capability.

To investigate the impact of spatial scale on classification performance, we evaluated the OA across three datasets by varying the window size w within the set $\{3, 5, 7, 9, 11\}$. The experimental results are illustrated in Fig. 7. It can be observed that as the window size increases from 3×3 to 9×9, the classification accuracy shows a steady upward trend across all three datasets. This improvement can be attributed to the effective utilization of local spatial coherence. However, when the window size is further increased to 11×11, the performance gains saturate or even slightly decline. This suggests that a 9×9 window achieves an optimal balance between capturing sufficient spatial context and avoiding the inclusion of irrelevant or interfering neighborhood information. Consequently, we set the spatial window size to 9×9 for all subsequent experiments on these datasets.

4) Embedding Dimension Analysis: An appropriate embedded dimension r is important for DR. If the dimension is too high, it will significantly increase the computational load. On the other hand, if the dimension is too low, key information will be lost. We analyzed the proposed t-LRDGE model with varying values of r, while keeping the experimental conditions identical to those in Section IV-A. From the results in Fig. 8, we can draw some conclusions about these DR methods on the three datasets.

When r is selected from the set of values {2, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50}, the OA of most DR methods initially increases and then remains stable or experiences a slight decline. The proposed t-LRDGE model demonstrates marked superiority over other methods under most of the embedding dimensions. Considering that all the DR methods reach peak values when $r > 30$, we set r as 30 for all the DR methods in the experiments.

5) Ablation Study Analysis: The ablation study is performed to illustrate the advantageous effects of the proposed three modules in the t-LRDGE model. For this study, we utilize Indian Pines, Salinas, and LongKou databases. The following experiments are carried out according to the same configuration as described in Section IV-A.

In order to verify the effectiveness of the GLR module, the LGE module, and the DE module, we set up a series of experiments by manually adjusting the tradeoff parameters λ_1, λ_2, and λ_3. Specifically, we first set the tradeoff parameter λ_2 corresponding to the DE module and the tradeoff parameter λ_3 corresponding to the LGE module to zero, respectively, so as to individually deactivate operations for these two modules. In addition, we deactivated the GLR module by removing the nuclear norm constraint $\|\mathcal{L}\|_*$ and the sparse term $\|\mathcal{E}\|_1$. The optimization formulas after deactivating the relevant term are shown in (50), (51), and (52), respectively, with their optimization steps referable to the original optimization

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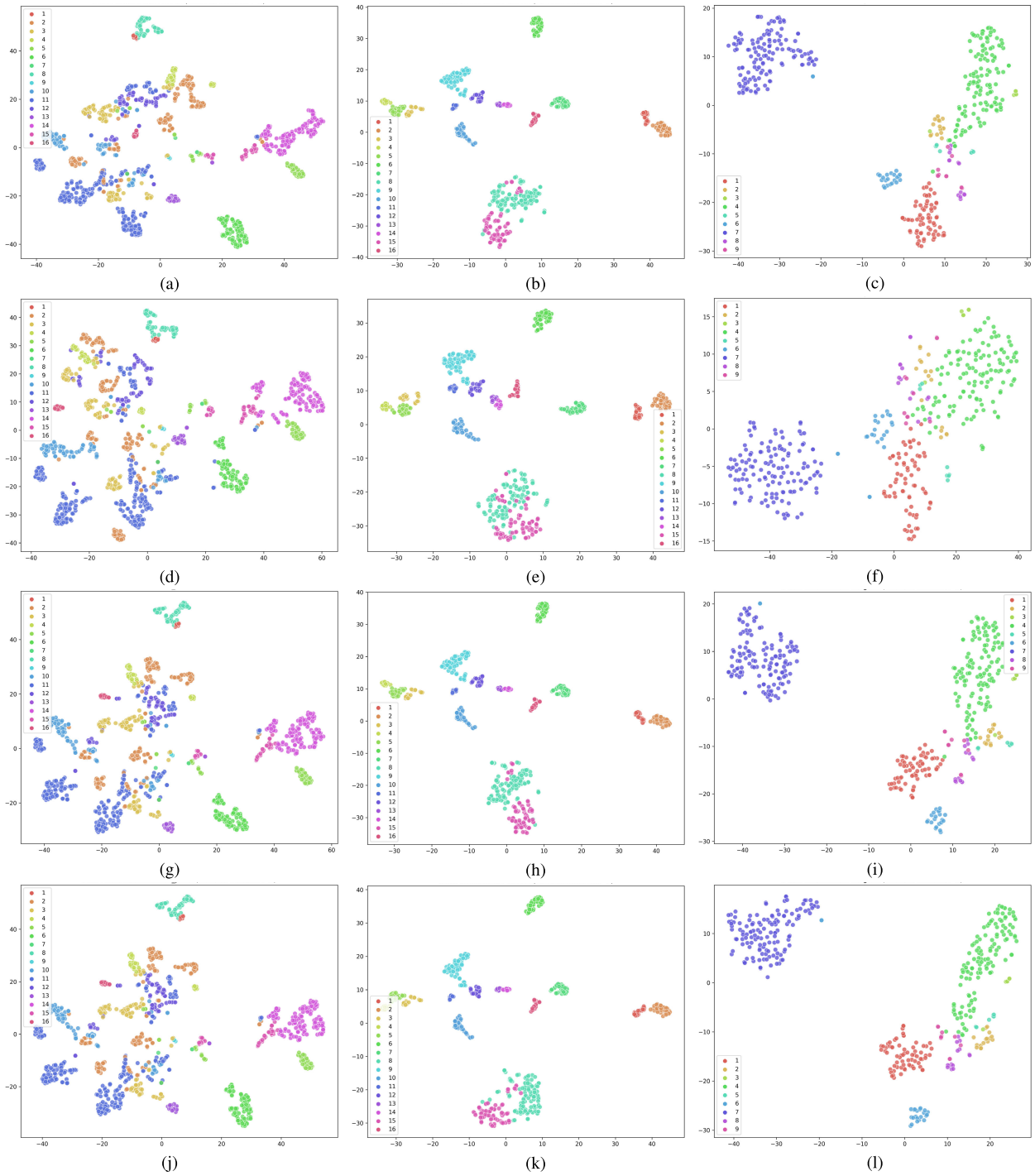


Fig. 9. t-SNE visualizations of the ablation study to verify the contribution of each module in t-LRDGE. (a) Indian Pines (w/o LGE). (b) Salinas (w/o LGE). (c) LongKou (w/o LGE). (d) Indian Pines (w/o DE). (e) Salinas (w/o DE). (f) LongKou (w/o DE). (g) Indian Pines (w/o GLR). (h) Salinas (w/o GLR). (i) LongKou (w/o GLR). (j) Indian Pines. (k) Salinas. (l) LongKou.

process of the t-LRDGE model detailed in Section III-C

$$\begin{aligned}
\min_{\mathcal{H},\mathcal{G},\mathcal{P},\mathcal{Q}} \quad & \lambda_2 \|\mathcal{Y}-\mathcal{G}*\mathcal{X}\|_F^2 \\
& + \lambda_3 \sum_i \sum_j \left\|\mathcal{Q}*\mathcal{X}_i-\mathcal{Q}*\mathcal{X}_j\right\|_F^2 W_{ij} \\
\text{s.t.} \quad & \mathcal{P}'*\mathcal{P}=\mathcal{I}, \quad \mathcal{H}*\mathcal{P}=\mathcal{G}, \quad \mathcal{P}=\mathcal{Q} \\
& \sum_i \|\mathcal{Q}*\mathcal{X}_i\|_F^2 \; d_{ii}=1
\end{aligned} \tag{50}$$

$$\begin{aligned}
\min_{\mathcal{M},\mathcal{L},\mathcal{E},\mathcal{P},\mathcal{Q}} \quad & \|\mathcal{M}\|_* + \lambda_1 \|\mathcal{E}\|_2 \\
& + \lambda_3 \sum_i \sum_j \left\|\mathcal{Q}*\mathcal{X}_i-\mathcal{Q}*\mathcal{X}_j\right\|_F^2 W_{ij} \\
\text{s.t.} \quad & \mathcal{P}*\mathcal{X}=\mathcal{P}*\mathcal{L}+\mathcal{E}, \quad P'*P=\mathcal{I} \\
& \mathcal{L}=\mathcal{M}, \quad \mathcal{P}=\mathcal{Q} \\
& \sum_i \|\mathcal{Q}*\mathcal{X}_i\|_F^2 \; d_{ii}=1
\end{aligned} \tag{51}$$

$$\min_{\mathcal{M},\mathcal{L},\mathcal{E},\mathcal{H},\mathcal{G},\mathcal{P}} \quad \|\mathcal{M}\|_* + \lambda_1 \|\mathcal{E}\|_2 + \lambda_2 \|\mathcal{Y}-\mathcal{G}*\mathcal{X}\|_F^2$$
$$\text{s.t.} \quad \mathcal{P}*\mathcal{X}=\mathcal{P}*\mathcal{L}+\mathcal{E}, \quad P'*P=\mathcal{I}$$

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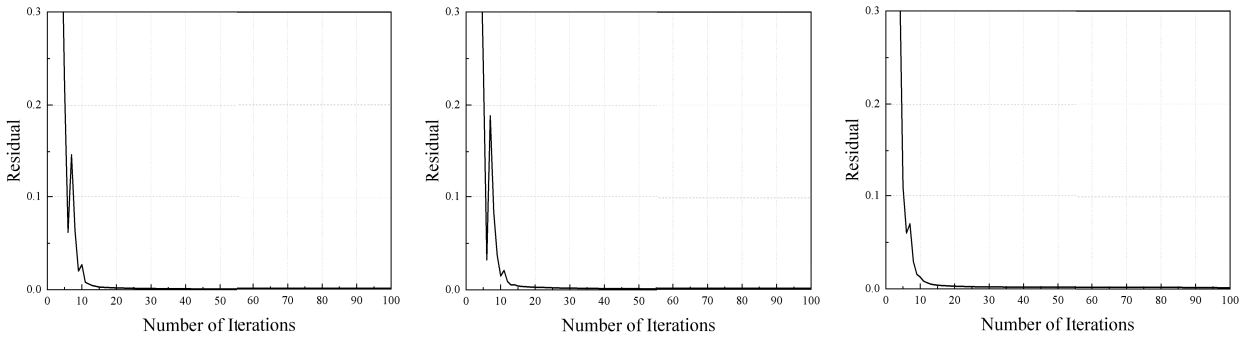


Fig. 10. Empirical convergence curves of t-LRDGE. (a) Indian Pines. (b) Salinas. (c) LongKou.

TABLE V
ABLATION STUDY ON THE LGE, DE, AND GLR MODULE

LGE	DE	GLR	Indian	Salinas	Longkou
×	✓	✓	96.40 (-1.1)	97.21 (-1.25)	97.35 (-0.39)
✓	×	✓	93.93 (-3.57)	92.90(-5.56)	89.54(-8.2)
✓	✓	×	97.22 (-0.28)	98.19(-0.27)	97.20(-0.54)
✓	✓	✓	97.50	98.46	97.74

$$\mathcal{L} = \mathcal{M} \quad \mathcal{H} * \mathcal{P} = \mathcal{G}. \tag{52}$$

The relevant experimental results are presented in Table V. It is clear that the performance of the t-LRDGE model significantly degrades when any of the three modules is removed, and the DE module has a more pronounced impact on classification accuracy. This is because the DE module provides effective semantic embedding. By narrowing the gap between the low-dimensional data space and the label space, it directly promotes a substantial improvement in classification accuracy. Specifically, on Indian, Salinas, and LongKou datasets, the classification accuracy decreases by 1.1%, 1.25%, and 0.39%, respectively, after removing the GE module; while the removal of DE leads to accuracy reductions of 3.57%, 5.56%, and 8.2%, respectively. Furthermore, removing the GLR module results in accuracy drops of 0.08%, 0.27%, and 0.54%, respectively. The above experimental results fully demonstrate that the GLR, LGE, and DE modules work together to reach an optimal solution.

The t-SNE visualizations in Fig. 9 reveal that the absence of the DE or LGE module leads to less compact feature clustering and significantly reduced separability. Although the GLR module shows limited contribution to accuracy on clean data, its primary value lies in enhancing the robustness against noise in complex environments (as detailed in Section IV-C1). Overall, the full t-LRDGE model effectively balances feature discriminability and robustness, achieving the best clustering performance.

6) Convergence Analysis: To investigate the convergence behavior of the proposed t-LRDGE model, we recorded the objective function values generated by Algorithm 2 during the optimization process. Fig. 10 illustrates the convergence curves on Indian Pines, Salinas, and LongKou datasets. It can be observed that the objective function values decrease monotonically and drop rapidly in the first few iterations, stabilizing within approximately 50 iterations. This empirical evidence validates the effectiveness and efficient convergence of our ADMM-based iterative optimization strategy.

V. CONCLUSION

In this article, we propose a t-LRDGE method for HSI DR. The t-LRDGE integrates the ability of t-product operations to preserve spatial structure in tensor representations and transformations, the global information capture and noise robustness of low-rank decomposition, the local geometric information preservation advantage of GE, and the precise DE property of semantic priors, forming a unified DR framework for joint optimization. It not only provides a new scheme for HSI DR that balances structural integrity and information utilization, but also offers a novel perspective for the deep fusion of tensor computing and projection learning. However, we also acknowledge certain limitations of the proposed method. Specifically, compared to closed-form matrix-based approaches, the iterative ADMM optimization involving tensor operations entails higher computational complexity. In addition, the sensitivity to multiple tradeoff parameters necessitates a time-consuming grid search for optimal tuning. In future work, we will focus on optimizing the computational efficiency of the model and investigating more efficient numerical optimization strategies to reduce the complexity of tensor iterative computations.

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